# INDUCED PATHS, RADIUS, AND EXTREMAL NEIGHBORHOOD INDEPENDENCE IN GRAPHS WITHOUT 4-CYCLES

XINGCHEN WANG AND NEIL XU

ABSTRACT. Let $M(n)$ be the maximum of $\ell(G) = S(G)/n$ over connected graphs of order $n$ with no 4-cycle subgraph and no induced path on six vertices, where $S(G)$ is the sum of $\alpha(G[N(v)])$. Then $M(7) = 16/7$, $M(8) = 5/2$, $M(9) = 8/3$, $M(10) = 3$ and $M(n) = 2 + 2/n$ for every $n \geq 11$, and for every $n \geq 7$ the extremal graph is unique; in particular $S \leq 3n$, with equality only for the Petersen graph. A connected graph with no 4-cycle subgraph, of radius 4 and induced-path order 7, is $C_8$ or a 6-cycle extended by a matching and isolated vertices at each cycle vertex and at most one further vertex on each cycle edge, and satisfies $S = 2n - b - m$. For connected triangle-free graphs with no 4-cycle subgraph and no induced path on seven vertices, $2e \leq 3n$ with equality only for the Petersen graph, and the maximum of $2e$ on $n \geq 10$ vertices is $3n - \min\{n - 10,\, 5 + \mathbf{1}_{\{n \text{ even}\}}\}$; a graph in the same path-and-cycle class with exactly one triangle still satisfies $S \leq 3n$. Every connected graph with no 4-cycle subgraph satisfies $p \geq r + \ell$ whenever $\ell \leq 3$, whenever $r \leq 2$ and $\ell \leq 4$, whenever $r = 4$ and $\ell \leq 4$, and whenever $r \geq 5$ and $\ell \leq 4$. Every remaining counterexample to Conjecture 133 of *Written on the Wall II* with $\ell \leq 4$ lies in the cell $r = 3$, $p = 6$, $T \geq 2$, $n \geq 17$ and $3 < \ell \leq 4 - 16/n$.

## 1. INTRODUCTION

All graphs are finite and simple. For a graph $G$ with $n \geq 2$ vertices let $p(G)$ be the largest number of vertices of an induced path, $r(G)$ the radius, and

$$\ell(G) = \frac{1}{n} \sum_{v \in V(G)} \alpha\big(G[N(v)]\big),$$

where $\alpha$ is the independence number and $N(v)$ the open neighbourhood. Write $S(G)$ for the sum in the numerator, and write $c_{C_4}(G) = 1$ if $G$ contains no 4-cycle as a subgraph and $c_{C_4}(G) = 0$ otherwise. A graph with no 4-cycle subgraph is called *strong $C_4$-free*.

The two structural results of the paper are Theorems 3.6 and 4.3. Let $P$ denote the Petersen graph and $P - v$ the graph obtained by deleting one vertex. Theorem 3.6 determines the maximum $M(n)$ of $\ell$ over connected strong $C_4$-free $P_6$-free graphs of order $n \geq 7$. The value is $16/7$ at order 7, $5/2$ at order 8, attained only by the graph $H_8$ of Figure 1, $8/3$ at order 9, attained only by $P - v$, and 3 at order 10, attained only by $P$. For every $n \geq 11$ the maximum is $2 + 2/n$. For every $n \geq 7$, the extremal graph is unique up to isomorphism. The same theorem gives $S \leq 3n$, with equality only for $P$.

The shape of the extremal graph changes between orders 10 and 11, and the change is forced by the induced 6-cycle branch. A vertex off such a cycle meets it in an antipodal pair. Distinct off-cycle vertices of the first distance layer use distinct pairs, so there are at most three of them. If at least two antipodal pairs are occupied, any two vertices of the second distance layer would have two common neighbours in the first layer. That layer therefore has size at most one, and the order is $6 + k + h \leq 10$. The graph of order 10 in this branch is $P$. At order 11 the branch with two or more occupied pairs is empty. Every remaining graph in the induced 6-cycle branch has a single off-cycle vertex $a$ joined to one antipodal pair, and the second distance layer induces

*Date*: November 2026.

*Key words and phrases.* induced path, radius, neighbourhood independence, 4-cycle, WOWII Conjecture 133.

a matching together with isolated vertices. If that matching has $m$ edges, the count in the proof of Theorem 3.6 gives $S = 2n + 2 - m$. Equality $S = 2n + 2$ holds only for $m = 0$, that is, only when every vertex of the second layer is a leaf, and the extremal graph is then the graph $\Gamma_{n-7}$ of Figure 1. The class itself is not a single graph: the graph obtained from $\Gamma_4$ by joining two of its leaves is strong $C_4$-free and $P_6$-free of order 11, and it has $S = 23$, strictly below the extremal value 24. Every graph in the induced 5-cycle branch has $S \le 2n + 1$, and every chordal graph in the class has $S \le 2n - 2$, so both fall short of $2n + 2$. Uniqueness of the extremal graph can fail for smaller orders. The same case division shows that at order 4 both $P_4$ and $K_{1,3}$ attain $S = 6$, and that at order 6 both $C_6$ and the graph obtained from $C_5$ by adding one leaf attain $S = 12$.

An enumeration of the connected strong $C_4$-free $P_6$-free graphs of order at most 12 agrees with this uniqueness statement. It is not used in the proofs.

| $n$ | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|
| number of graphs | 43 | 87 | 183 | 355 | 699 | 1327 |
| maximum of $S$ | 16 | 20 | 24 | 30 | 24 | 26 |
| number of extremal graphs | 1 | 1 | 1 | 1 | 1 | 1 |

Theorem 4.3 describes every connected strong $C_4$-free graph of radius 4 and induced-path order 7. Fajtlowicz characterised the radius-critical graphs, and DeLaViña, Larson, Pepper and Waller record the consequence used here [9, 8]. An $r$-ciliate is a graph $C(2t, r - t)$ for some $1 \le t \le r$: one endpoint of each of $2t$ paths of order $r - t + 1$ is joined in a cycle, with the extreme case $t = 1$ read as the path $P_{2r}$ and the extreme case $t = r$ read as $C_{2r}$. The four 4-ciliates are therefore $P_8$, the 4-cycle with a pendant path of length 2 at each vertex, the 6-cycle with a pendant edge at each vertex, and $C_8$. The path $P_8$ is excluded by the hypothesis $p = 7$: that path has induced-path order 8, and it has no 4-cycle core. The 4-cycle with pendant paths of length 2 is excluded because the core is a 4-cycle subgraph. The two skeletons that remain are $C_8$ and the 6-cycle with a pendant leaf at each vertex. The new content is the classification of every extension of that skeleton. The sets $C_i$ of vertices with trace $\{v_i\}$ may be nonempty simultaneously, the only edges off the cycle are a matching inside a single $C_i$ or a vertex glued to one cycle edge, and $S = 2n - b - m$, where $b$ is the number of glued vertices and $m$ is the number of matching edges. The device that forbids an edge with ends in distinct sets $C_i$ is a distance-4 witness $w_i \in C_i$: such an edge completes an induced $P_8$ through two of these witnesses. Figure 2 draws that path, and on the same cycle it draws an isolated leaf, a matching edge and a glued vertex, so that the contributions to $b + m$ are visible. Corollaries 4.4 and 4.5 give $S$ and the equality case. Neither theorem is a special case of the conjecture recorded next.

**Conjecture 1.1** (Graffiti/WOWII Conjecture 133)**.** *Every connected graph $G$ on at least two vertices satisfies*

$$r(G) + \ell(G)^{\,c_{C_4}(G)} \le p(G).$$

The conjecture is number 133 in E. DeLaVina's list *Written on the Wall II* of conjectures of the program Graffiti.pc [7]. We follow the formal statement in the Formal Conjectures repository at the fixed commit `5faef0c4dc3d226f8f9a9fc8f581b6221283eb6e` [11], the merge commit of pull request #5727 [12]. That commit removed the floor function from an earlier version of the file. The statement there reads `(rad : ℝ) + l G ^ cC4 ≤ (path G : ℝ)`. We work with this real-valued statement and call it REAL-133.

If $G$ contains a 4-cycle, the conjecture reads $p(G) \ge r(G) + 1$. A shortest path between two vertices at distance $\operatorname{diam}(G) \ge r(G)$ is induced, so the inequality holds. Only strong $C_4$-free graphs remain. Since $p$ and $r$ are integers, REAL-133 for such a graph is equivalent to

$$p(G) - r(G) \;\ge\; \lceil \ell(G) \rceil. \tag{1}$$

Theorem 5.13 is the range in which the estimates of this paper give REAL-133. The triangle-free bound is Theorem 5.4, and Theorem 5.5 is the exact value of that maximum for every order $n \geq 10$. With the deficit lemmas of Section 5 and the one-triangle bound of Theorem 5.6, every counterexample in the range $\ell \leq 4$ falls in the cell of Corollary 5.11: radius 3, induced-path order 6, at least two triangles, order at least 17, and $3 < \ell \leq 4 - 16/n$.

H. Chen [3] studies the floor form $p \geq r + \lfloor \ell \rfloor$ of the same conjecture. The arguments below do not use his proofs. His Lemma 2.1 is the local matching lemma: in a strong $C_4$-free graph the neighbourhood of every vertex is a disjoint union of edges and isolated vertices, and $S = 2e - 3T$. Lemma 2.1 records the same description. His Lemma 7.18 states that if an induced 6-cycle lies in a strong $C_4$-free $P_6$-free graph, then every vertex outside the cycle has either no neighbour on the cycle or exactly two, and those two are antipodal. His Corollary 7.20 records the consequence for the cycle vertices: each has triangle count zero, and its degree is 2 or 3 according as its antipodal slot is empty or occupied. The same restriction on traces is Lemma 3.4 when the cycle dominates, and it is the description of the first distance layer in the proof of Theorem 3.6. His Theorem 6.7 states that a connected $C_4$-subgraph-free graph of diameter 2 with $\ell > 3$ has an induced path on at least six vertices. On the slice $r = 2$ and $3 < \ell \leq 4$ this is the same numerical conclusion as Theorem 5.13. The increment here is that Theorem 5.13 also covers radius 2 when the diameter is larger than 2, while Theorem 3.6 determines the extremal value at every order. Immediately after the proof of his Theorem 3.6, Chen records that every regular graph of girth at least 5 and radius 2 satisfies the floor form, and that this includes every Moore graph of diameter 2. A Moore graph of diameter 2 has $\ell$ equal to its degree, an integer, so the floor form and the real form coincide. The Moore assertion of Proposition 2.4 in the supplement is an alternate proof of that known case.

The exact function $M(n)$ is not a consequence of those lemmas. The proof of Theorem 3.6 continues to the second distance layer, and it treats the induced 5-cycle branch and the chordal branch separately. For every $n \geq 7$ it also gives uniqueness. Below that range the same division produces two extremal graphs at orders 4 and 6, as recorded above. Theorem 5.4 is the two-step argument from the classification, due to Chen and Xu [5, Lemma 3.4], of the connected triangle-free strong $C_4$-free $P_7$-free graphs with no clique cutset. If $v$ is a cut vertex, induced paths that start at $v$ in distinct components of $G - v$ concatenate, so one side has rooted path order at most 3 and falls under Lemma 5.3. If there is no cut vertex but $\{u, v\}$ is a clique cutset, a shortest $u$–$v$ path in each of two components has three internal vertices, and those six vertices together with $v$ induce a $P_7$. Restoring clique cutsets therefore yields no new equality graph: every graph with a clique cutset is strictly below $2e = 3n$, and the only equality graph on the Chen–Xu list is the Petersen graph.

Examples 3.1 and 3.7 in the supplement start from an $r$-ciliate of larger radius and add further blocks. They are not part of the radius-4 classification. In Example 3.1 each added path of length four contributes 3 vertices and 4 edges, so it increases $S = 2e$ by 8 while the radius and the induced-path order remain those of the ciliate. In Example 3.7 each attached copy of the incidence graph of a projective plane adds $(q + 1)f$ to $S$ and $f - 1$ to the order, and the radius and the path order remain $t + 3$ and $2t + 5$.

Section 3 proves Theorem 3.6. Section 4 proves Theorem 4.3. Section 5 proves the triangle-free bound, the exact function $f(n)$, the one-triangle bound, and the strengthened residual cell. Section 6 is the application to REAL-133, including the $\theta_k$ examples and the cells in the ranges $4 < \ell \leq 5$ and $5 < \ell \leq 6$. The rooted inequalities, the endpoint obstructions, the high-radius constructions, and the conditional reduction for $\mathrm{GEN}_3$ are in the supplement; they are not used for Theorems 3.6, 4.3, 5.4 and 5.5 or for Corollary 5.11. Table 1 is the list of what is proved and what is open. A finite search is not a proof in this article. Where the repository does not contain

a versioned checker, an input list, and a completeness argument, the corresponding statement is not used as a hypothesis.

## 2. Preliminaries

For $v \in V(G)$ let $d(v)$ be its degree and $t(v)$ the number of triangles containing $v$. Write $\lambda(v) = \lambda_G(v) = \alpha(G[N(v)])$, $S(G) = \sum_v \lambda(v)$, $e(G)$ for the number of edges and $T(G)$ for the number of triangles.

**Lemma 2.1.** *Let $G$ be strong $C_4$-free. Then:*

1. *two distinct vertices have at most one common neighbour;*
2. *for every $v$, $G[N(v)]$ is a matching together with isolated vertices, so $\lambda(v)$ equals the number of components of $G[N(v)]$ and $\lambda(v) = d(v) - t(v)$;*
3. *$S(G) = 2e(G) - 3T(G)$.*

*Proof.* (1) Two common neighbours $w, w'$ of $x, y$ give the 4-cycle $xwyw'$. (2) A vertex of $N(v)$ with two neighbours in $N(v)$ would have two common neighbours with $v$. So the components are single vertices and edges, each edge being a triangle at $v$. (3) Sum $d(v) - t(v)$ over $v$. □

The components of $G[N(v)]$ are the *units* of $v$. For a vertex set $U$, $G-U$ is the subgraph induced by $V \setminus U$, and $D_G(U) = S(G) - S(G - U)$ is the *deletion loss* of $U$. We write $L(v) = D_G(\{v\})$.

**Lemma 2.2.** *Let $G$ be strong $C_4$-free and $\emptyset \neq U \subseteq V(G)$. Then*

$$D_G(U) = \sum_{u \in U} \lambda(u) + \sum_{x \notin U} \#\{\text{units of } x \text{ contained in } U\}.$$

*In particular $L(v) = 2\lambda(v) - t(v) = 2d(v) - 3t(v)$.*

*Proof.* For $x \notin U$, deleting $U$ removes exactly the units of $x$ that lie inside $U$; a unit that is an edge with one end in $U$ survives as a vertex. For $U = \{v\}$: the $2t(v)$ neighbours of $v$ in triangles with $v$ keep their value, and each of the other $d(v) - 2t(v)$ neighbours loses one. The total is $\lambda(v) + d(v) - 2t(v) = 2\lambda(v) - t(v)$. □

**Lemma 2.3.** *Let $v$ be a cut vertex of $G$. Split the components of $G - v$ into two nonempty groups, and let $G_1, G_2$ be the subgraphs induced by the two groups together with $v$. Then $S(G) = S(G_1) + S(G_2)$.*

*Proof.* $G[N(v)]$ has no edges between the groups, so $\lambda_G(v) = \lambda_{G_1}(v) + \lambda_{G_2}(v)$. Every other vertex keeps its neighbourhood. □

We use three published results.

**Theorem 2.4** (Erdős, Saks and Sós [10])**.** *Every connected graph $G$ has an induced path on at least $2r(G) - 1$ vertices.*

**Theorem 2.5** (Camby and Schaudt [1, Theorem 5])**.** *Let $k \geq 4$ and let $G$ be a connected graph with no induced path on $k$ vertices. Then $G$ has a connected dominating set $X$ such that $G[X]$ has no induced path on $k - 2$ vertices or $G[X]$ is isomorphic to the cycle $C_k$.*

**Theorem 2.6** (Seinsche [14])**.** *If a graph on at least two vertices has no induced path on four vertices, then it or its complement is disconnected.*

Theorem 2.5 is the implication (i)⇒(ii) of [1, Theorem 5] applied with $H = G$. We write $P_k$ for the path on $k$ vertices, and "$G$ is $P_k$-free" means that $G$ has no induced $P_k$, that is, $p(G) \leq k - 1$. A connected graph of radius $r$ has at least $2r$ vertices.

**Lemma 2.7.** *Let $H$ be a connected strong $C_4$-free $P_4$-free graph on at least two vertices. Then $H$ has a vertex adjacent to all other vertices.*

*Proof.* By Theorem 2.6, $H$ is the join of two nonempty vertex sets $A$ and $B$. If $|A|, |B| \geq 2$, then $a, a' \in A$ and $b, b' \in B$ span the 4-cycle $aba'b'$. So one side is a single vertex, and it is adjacent to everything. □

## 3. Small-path mass bounds

**Lemma 3.1.** *If $G$ is connected and strong $C_4$-free with $r(G) = 1$, then $S(G) = 2(n-1) - m$, where $m$ is the number of edges of $G[N(u)]$ for a dominating vertex $u$. Hence $\ell(G) < 2$, and (1) holds.*

*Proof.* Let $v \neq u$. By Lemma 2.1(1), $v$ has at most one neighbour other than $u$, and such a neighbour lies in $N(u)$. So $G[N(v)]$ is a clique and $\lambda(v) = 1$; also $\lambda(u) = (n-1) - m$. Hence $S = 2(n-1) - m < 2n$ and $\lceil \ell \rceil \leq 2$. If $G$ is complete then $G \in \{K_2, K_3\}$, $\ell = 1$ and $p = 2 = r + 1$. Otherwise two nonadjacent vertices of $N(u)$ together with $u$ form an induced $P_3$, so $p \geq 3 \geq r + \lceil \ell \rceil$. □

**Lemma 3.2.** *If $G$ is connected and strong $C_4$-free with $r(G) \geq 2$, then $p(G) \geq 4$.*

*Proof.* If $G$ were $P_4$-free, Lemma 2.7 would give a dominating vertex, so $r(G) \leq 1$. □

**Lemma 3.3.** *If $G$ is connected, strong $C_4$-free and $P_5$-free, then $S(G) \leq 2n - h$, where $h$ is the number of vertices $v$ with $\lambda(v) \geq 3$. In particular $S(G) \leq 2n$.*

*Proof.* Call $v$ *simplicial* if $\lambda(v) = 1$ and *high* if $\lambda(v) \geq 3$. For $x \in N(v)$ put $X_x = N(x) \setminus N[v]$; then $N(x) \subseteq \{v\} \cup \{\bar{x}\} \cup X_x$, where $\bar{x}$ is the partner of $x$ if $x$ lies in an edge unit of $v$.

*Key fact.* Let $x, y$ lie in different units of $v$. A common vertex of $X_x$ and $X_y$ would give the 4-cycle $vxwy$, so $X_x \cap X_y = \emptyset$. If $x' \in X_x$ and $y' \in X_y$ are nonadjacent, then $x'xvyy'$ is an induced $P_5$: $x \not\sim y$ because they lie in different units, $x \not\sim y'$ and $y \not\sim x'$ by disjointness, and $v$ is adjacent to neither $x'$ nor $y'$. So $X_x$ is complete to $X_y$. If both are nonempty and $|X_x| \geq 2$, a vertex of $X_y$ has two neighbours in $X_x$, which are also neighbours of $x$, a contradiction to Lemma 2.1(1). Hence if both are nonempty, then $|X_x| = |X_y| = 1$.

*Main step.* For each high $v$ we find $\lambda(v) - 1$ simplicial vertices whose only high neighbour is $v$, one in each of $\lambda(v) - 1$ distinct units of $v$.

Case 1: some $u \in N(v)$ has $|X_u| \geq 2$. By the key fact $X_y = \emptyset$ for every $y$ outside the unit of $u$. Then $N(y) \subseteq \{v, \bar{y}\}$, which is a clique, so $y$ is simplicial; so is $\bar{y}$. Hence $v$ is the only high neighbour of $y$, and the $\lambda(v) - 1$ units other than that of $u$ supply the vertices.

Case 2: $|X_x| \leq 1$ for all $x \in N(v)$. Then $\lambda(x) \leq 2$ for $x \in N(v)$. Call a unit *active* if one of its vertices $x$ has $X_x \neq \emptyset$. Suppose two units are active, with $y' \in X_y$ and $z' \in X_z$, and let $x$ lie in a third unit. By the key fact $y' \sim z'$, and $xvyy'z'$ is an induced $P_5$: $x \not\sim y$ (different units), $x \not\sim y', z'$ (disjointness), $v \not\sim y', z'$, and $y \not\sim z'$ (disjointness). As $\lambda(v) \geq 3$, a third unit exists, so at most one unit is active. Every vertex of an inactive unit is simplicial and its only high neighbour is $v$. This gives at least $\lambda(v) - 1$ units.

*Counting.* The chosen vertices for distinct high vertices are distinct, since each has a unique high neighbour. So $\sum_{v \text{ high}} (\lambda(v) - 1) \leq \#\{\text{simplicial vertices}\}$, and

$$S = \sum_v \lambda(v) \leq 2n - \#\{\text{simplicial}\} + \sum_{v \text{ high}} (\lambda(v) - 2) \leq 2n - h.$$ □

**Lemma 3.4.** *Let $G$ be connected, strong $C_4$-free and $P_6$-free with a dominating induced cycle $C = v_0 v_1 \cdots v_5$, and let $W = V(G) \setminus V(C)$. Then every $w \in W$ is adjacent to exactly one antipodal pair $\{v_i, v_{i+3}\}$ and to no other vertex of $C$, distinct vertices of $W$ use distinct pairs, and $W$ is independent. Hence $|W| \leq 3$, $S(G) = 12 + 4|W| < 3n$, and $r(G) \leq 2$ if $W \neq \emptyset$.*

*Proof.* Indices are modulo 6. Let $w \in W$; it has a neighbour on $C$. If $w$ has exactly one neighbour $v_i$, then $wv_iv_{i+1}v_{i+2}v_{i+3}v_{i+4}$ is an induced $P_6$. If $N(w) \cap C = \{v_i, v_{i+1}\}$, then $wv_{i+1}v_{i+2}v_{i+3}v_{i+4}v_{i+5}$ is an induced $P_6$. Two neighbours at distance 2 on $C$ give a 4-cycle. A set of at least three vertices of $C_6$ contains two vertices at distance 2. So $N(w) \cap C$ is an antipodal pair. Two vertices of $W$ with the same pair give two common neighbours. If $w \sim w'$ with distinct pairs, some $v_i \in N(w)$ and $v_j \in N(w')$ are adjacent on $C$, and $wv_iv_jw'$ is a 4-cycle. There are no triangles, $e(G) = 6 + 2|W|$, and Lemma 2.1(3) gives $S = 12 + 4|W|$. Finally, if $w$ uses $\{v_i, v_{i+3}\}$, then every vertex of $C$ is within distance 2 of $v_i$ (the vertex $v_{i+3}$ through $w$), and each $w' \in W$ has a neighbour in $\{v_{i-1}, v_i, v_{i+1}\}$, because each of the three antipodal pairs meets this set. So $v_i$ has eccentricity 2. □

The same trace analysis applies to any vertex adjacent to an induced 6-cycle in a connected strong $C_4$-free $P_6$-free graph, whether or not the cycle dominates. Theorem 3.6 uses that form.

**Lemma 3.5.** *Let $G$ be connected, strong $C_4$-free and $P_6$-free. Then either $G$ has a vertex of eccentricity at most 2, or $G = C_6$.*

*Proof.* Apply Theorem 2.5 with $k = 6$ to get a connected dominating set $X$. If $G[X] \cong C_6$, Lemma 3.4 gives $G = C_6$ or a vertex of eccentricity 2. Otherwise $G[X]$ is $P_4$-free. If $|X| = 1$, the vertex in $X$ dominates $G$. If $|X| \geq 2$, Lemma 2.7 gives $u \in X$ adjacent to all of $X \setminus \{u\}$, and since $X$ dominates $G$, every vertex is within distance 2 of $u$. □

**Theorem 3.6.** *Let $M(n)$ be the maximum of $\ell(G)$ over connected strong $C_4$-free $P_6$-free graphs of order $n \geq 7$. Then*

$$M(n) = \begin{cases} 16/7 & \text{if } n = 7, \\ 5/2 & \text{if } n = 8, \\ 8/3 & \text{if } n = 9, \\ 3 & \text{if } n = 10, \\ 2 + 2/n & \text{if } n \geq 11. \end{cases}$$

*For every $n \geq 7$ the extremal graph is unique up to isomorphism. For $n = 10$ it is the Petersen graph $P$. For $n = 9$ it is $P - v$. For $n = 8$ it is the graph $H_8$ of Figure 1. For $n = 7$ and for every $n \geq 11$ it is the graph $\Gamma_{n-7}$ of that figure: a 6-cycle, a vertex $a$ adjacent exactly to one antipodal pair, and $n - 7$ leaves adjacent only to $a$. In particular, every connected strong $C_4$-free $P_6$-free graph $G$ of order $n \geq 2$ satisfies $S(G) \leq 3n$, with equality if and only if $G$ is the Petersen graph. At orders 4 and 6 the maximum is attained by two nonisomorphic graphs.*

*Proof.* Throughout, $S = 2e - 3T$ by Lemma 2.1. A chordless cycle of length at least seven contains an induced $P_6$. Thus the only possible holes have lengths five and six.

*Case 1: an induced six-cycle.* Label its vertices $c_0, \ldots, c_5$, indices modulo 6, and write $A$ for the set of vertices at distance one from the cycle. The trace argument in the proof of Lemma 3.4 shows that each $a \in A$ meets the cycle in one antipodal pair. Distinct vertices of $A$ use distinct pairs, and $A$ is an independent set: an edge of $A$ joins distinct pairs, and some contact of one end is adjacent on the cycle to some contact of the other, giving a 4-cycle. Hence $k = |A| \leq 3$. Figure 1 shows the three extremal graphs of this branch.

Every vertex $x$ at distance two from the cycle is adjacent to every vertex of $A$. Such an $x$ has at least one neighbour in $A$. If $x \sim a$ and $x \not\sim b$ for $a, b \in A$, choose contacts so that $c_i \in N(a)$ and $c_{i+2} \in N(b)$. This is possible: the pairs are distinct, so some contact of $b$ lies at cycle distance two from some contact of $a$, and we orient the cycle from that contact of $a$ toward that contact of $b$. The path $x, a, c_i, c_{i+1}, c_{i+2}, b$ is then an induced $P_6$. Indeed $x$ meets the cycle in nothing, $a$ meets the cycle only in its antipodal pair (neither $c_{i+1}$ nor $c_{i+2}$), $b$ meets the cycle only in its antipodal pair (so $b \sim c_{i+2}$ and $b \not\sim c_i, c_{i+1}$), and $a \not\sim b$.

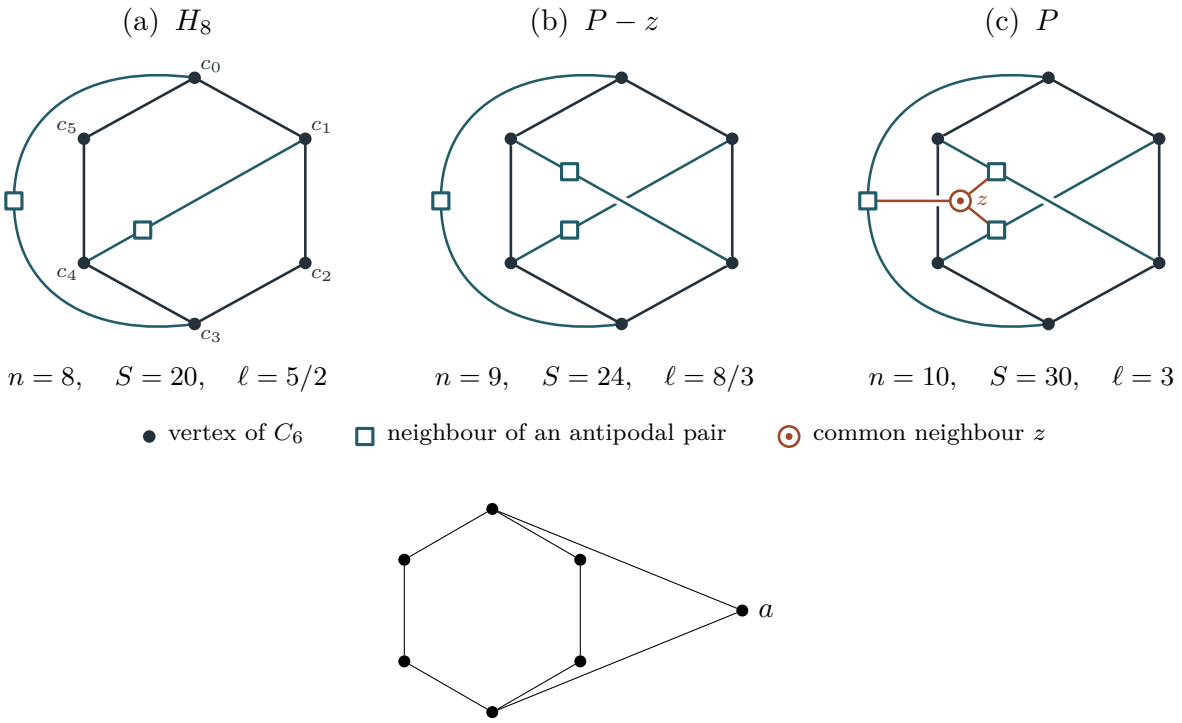



FIGURE 1. Extremal graphs for $M(n)$. In $H_8$, two antipodal pairs on the 6-cycle have a common neighbour outside the cycle. Adding a common neighbour for the third pair gives $P - z$. Joining a new vertex $z$ to the three outside vertices gives the Petersen graph $P$. The three graphs are triangle-free, so $S = 2e$. The drawing underneath is $\Gamma_0$, the unique graph of order 7 with $\ell = 16/7$: the marked vertex $a$ is adjacent to one antipodal pair. Attaching any number of leaves at $a$ gives the unique extremal graph of each order $n \geq 11$, with $S = 2n + 2$.

There is no vertex at distance three or more from the cycle. A shortest path $z, x, a, c_i$ from such a vertex down to the cycle, extended by $c_{i+1}, c_{i+2}$, is an induced $P_6$: the vertex $z$ has no neighbour in $A$ or on the cycle, and $x$ has no neighbour on the cycle.

Let $h$ be the number of vertices at distance two. If $k = 0$, then $A$ is empty, so the cycle has no neighbour and $G = C_6$. If $k = 1$, every distance-two vertex is adjacent to the unique vertex $a \in A$. The distance-two vertices induce a matching together with isolated vertices: a vertex of that layer with two neighbours in the layer would have two common neighbours with $a$. Writing $m$ for the number of edges of the matching gives $n = 7 + h$, $e = 8 + h + m$ and $T = m$, so $S = 16 + 2h - m = 2n + 2 - m$. Thus $S \leq 2n + 2$, with equality if and only if $m = 0$. Equality means that $G$ is the graph $\Gamma_h$ formed by the 6-cycle, the vertex $a$ on one antipodal pair, and $h$ leaves at $a$.

The graph $\Gamma_h$ lies in the class, for every $h \geq 0$. A leaf has degree one. Every other cycle through an outside vertex uses both edges from $a$ to the cycle, and each of the two $a$-paths along the cycle has length three, so every such cycle has length five. There is no 4-cycle. There is no induced $P_6$. A leaf, having degree one, occupies an end of any path that contains it, and that path meets $a$ next. Two leaves therefore determine a path on only three vertices. An induced path that contains one leaf and five further vertices would restrict to an induced path of order five in $\Gamma_0$ that starts at $a$. No such path exists: the two neighbours of $a$ on the cycle are an antipodal pair, the path $a, c_i, c_{i+1}, c_{i+2}, c_{i+3}$ has the chord $ac_{i+3}$, and every shorter continuation from $a$ through

one neighbour stops at the vertex before the antipode, because the next cycle vertex is adjacent to $a$. An induced path of $\Gamma_0$ that avoids $a$ lies on the 6-cycle and has order at most five. An induced path of $\Gamma_0$ that contains $a$ in its interior has $c_i$ and $c_{i+3}$ as the two neighbours of $a$ on the path. Each side can then take at most one further cycle vertex: the next vertex on either arc is adjacent to the antipode already used. The resulting path has order at most five. Thus $\Gamma_h$ is strong $C_4$-free and $P_6$-free, and it attains $S = 2n + 2$.

If $k \in \{2, 3\}$, then $h \leq 1$: two vertices of the second layer would both be adjacent to two vertices of $A$. There are no triangles. The second-layer vertex, if it exists, is adjacent to every vertex of $A$ and to nothing on the cycle, $A$ is independent, and each vertex of $A$ contributes two edges to the cycle. Thus $n = 6 + k + h$ and $S = 2(6 + 2k + kh)$. The four possibilities are

| $(k, h)$ | $(2, 0)$ | $(2, 1)$ | $(3, 0)$ | $(3, 1)$ |
|---|---|---|---|---|
| $S/n$ | $5/2$ | $8/3$ | $8/3$ | $3$ |

The four pairs are the graphs of Figure 1, with $(3, 0)$ and $(2, 1)$ both equal to $P - z$. Write $a_i$ for the vertex adjacent to the antipodal pair $\{c_i, c_{i+3}\}$. The pair $(k, h) = (3, 1)$ is the graph in panel (c): the three vertices $a_0, a_1, a_2$ and a vertex $z$ adjacent to all three. It is cubic of order 10. It is triangle-free, and the trace rules give no 4-cycle, so its girth is at least 5. Its diameter is 2. Indeed every pair of cycle vertices is at distance at most 2 along the cycle or through the corresponding $a_i$; the vertex $z$ meets every $a_i$; and a non-neighbour of $a_i$ on the cycle is adjacent to $c_i$ or to $c_{i+3}$. Suppose $v_0 \cdots v_5$ were an induced path. Diameter 2 gives $v_0$ a common neighbour with each of $v_3, v_4, v_5$. Those neighbours lie off the path, because the only neighbour of $v_0$ on the path is $v_1$ and the path is induced. They are distinct: a vertex adjacent to two of $v_3, v_4, v_5$ would form a triangle with a path edge, or a 4-cycle with the length-2 subpath $v_3v_4v_5$. This is impossible: $v_0$ has degree 3, so it has only two neighbours off the path. Thus the graph is $P_6$-free. It is the Petersen graph $P$, and $\ell(P) = 3$.

Deleting $z$ leaves $(k, h) = (3, 0)$, the graph $P - z$ in panel (b). The pair $(2, 1)$ is isomorphic to the same graph. In the labelling with $A = \{a_0, a_1\}$ and second-layer vertex $z$, the cycle $c_0, a_0, c_3, c_4, a_1, c_1$ has its three antipodal pairs served by $c_5$, $z$ and $c_2$, and no vertex remains. Both graphs therefore have $\ell = 8/3$, and both are isomorphic to $P - v$. Induced subgraphs of $P$ are strong $C_4$-free and $P_6$-free. The pair $(2, 0)$ is the graph $H_8$ in panel (a), with $\ell = 5/2$. The three choices of the unused antipodal pair differ by a rotation, so this graph is unique up to isomorphism. Its order is 8, so it is neither $P$ nor $P - v$. For $k \in \{2, 3\}$ one has $n = 6 + k + h \leq 10$.

*Case 2: an induced five-cycle and no induced six-cycle.* Write $C = c_0 \cdots c_4c_0$. Every vertex at distance one from $C$ has a singleton or consecutive-pair trace: any other pair of contacts lies at distance two on $C$ and forms a 4-cycle. If a vertex $x$ were at distance two, with a neighbour $a$ in the first layer, the path $x, a$ followed by four consecutive cycle vertices, beginning at one contact of $a$ and omitting the second contact if $a$ has one, would be an induced $P_6$. Thus $C$ dominates $G$.

Write $h$ for the number of vertices with singleton trace and $j$ for the number with a consecutive-pair trace. At most one vertex carries each of the five pair traces, since two would have two common neighbours, so $j \leq 5$. The edges outside $C$ are of the following kinds.

- Two singleton vertices with different supports are nonadjacent. Neighbouring supports give a 4-cycle. Supports at distance two give an induced 6-cycle along the longer arc of $C$. Edges within one singleton class form a matching, because two neighbours in that class would be two common neighbours of the supporting cycle vertex. Let $m$ be the number of these edges.
- No two pair vertices are adjacent. From their two distinct support edges one can choose adjacent cycle vertices, one from each edge, and those two vertices together with the pair vertices form a 4-cycle.

- A singleton supported at $c_i$ meets only a pair vertex supported on the opposite edge $\{c_{i+2}, c_{i+3}\}$. Every other pair edge contains a neighbour of $c_i$ and yields a 4-cycle. That pair vertex meets at most one vertex of the singleton class, again by the common-neighbour bound. Let $b$ be the number of singleton-to-pair edges, so $b \le \min(h, j)$.

Consequently $n = 5 + h + j$ and $e = 5 + h + 2j + m + b$. The triangles are the $j$ pair triangles and the $m$ matching triangles, so $T = j + m$ and

$$S = 2e - 3T = 2n + 2b - j - m.$$

The integer $b$ is at most 1. The opposite edge of the support $c_i$ is $\{c_{i+2}, c_{i+3}\}$, and as $i$ runs through the cycle this edge runs through every edge exactly once. A singleton meets at most one vertex of the opposite pair class, and that class contains at most one vertex. Thus two singleton–pair edges use two distinct supports and the two opposite pair vertices. Supports at distance 1 are nonadjacent, for an edge between the corresponding singletons would form a 4-cycle with the cycle edge; supports at distance 2 are nonadjacent, for an edge between the corresponding singletons would form an induced 6-cycle along the longer arc. Pair vertices on distinct edges are nonadjacent, as already recorded.

Suppose the supports are at distance 2: a singleton $s$ at $c_0$ is adjacent to the pair vertex $p$ on $\{c_2, c_3\}$, and a singleton $t$ at $c_2$ is adjacent to the pair vertex $q$ on $\{c_4, c_0\}$. The path $s, p, c_2, t, q, c_4$ is an induced $P_6$. The consecutive non-edges are the trace restrictions. The distance-2 pairs on the path are $sc_2$, $pt$, $c_2q$ and $tc_4$: $s$ meets only $c_0$ on the cycle, $t$ meets only its opposite pair vertex $q$, $q$ meets the cycle only in $\{c_4, c_0\}$, and $t$ meets the cycle only in $c_2$. The distance-3 pairs are $st$, $pq$ and $c_2c_4$, all non-edges by the preceding paragraph or by the cycle. The distance-4 pairs are $sq$ and $pc_4$, and neither singleton–pair incidence is the opposite one.

Suppose the supports are at distance 1: $s$ at $c_0$ is adjacent to $p$ on $\{c_2, c_3\}$, and $t$ at $c_1$ is adjacent to $q$ on $\{c_3, c_4\}$. The path $s, p, c_3, q, t, c_1$ is an induced $P_6$, by the same trace restrictions: the only singleton–pair edges among these vertices are $sp$ and $tq$, the only cycle edges on the path are $pc_3$ and $qc_3$, and $t$ is adjacent to $c_1$ alone on the cycle. Either configuration contradicts the hypothesis that $G$ is $P_6$-free. Therefore $b \le 1$.

If $b = 0$, then $S = 2n - j - m \le 2n$. If $b = 1$, then $j \ge 1$ because $b \le \min(h, j)$, so $S = 2n + 2 - j - m \le 2n + 1$. In this case $S \le 2n + 1$.

*Case 3: no holes.* The graph is chordal, and every nonempty induced subgraph has a simplicial vertex. The neighbourhood of a simplicial vertex is a clique. The degree is at most two: four vertices of a clique contain a 4-cycle. A simplicial vertex of a connected chordal graph on at least two vertices is not a cut vertex, and the graph left by deleting it is chordal and strong $C_4$-free. In the graph present at the deletion, the loss is $L(v) = 2d(v) - 3t(v)$. Degree one gives loss 2. Degree two, with the two neighbours adjacent, gives one triangle and loss 1. Deleting vertices until a single vertex remains, one has $S \le 2(n - 1) = 2n - 2$.

The three cases give the values of $M(n)$. A graph of order $n \ge 11$ cannot realise $k \in \{2, 3\}$ in Case 1, and $k = 0$ forces $G = C_6$. The only remaining graphs in Case 1 are the graphs $\Gamma_h$ and the graphs with the same first layer and $m \ge 1$. The latter have $S \le 2n + 1$. Cases 2 and 3 give $S \le 2n + 1$ and $S \le 2n - 2$. Thus $M(n) = 2 + 2/n$, attained only by $\Gamma_{n-7}$.

At order 10 the only graph in the list with $S > 2 \cdot 10 + 2$ is $P$, with $S = 30$. At order 9 the only such graph is $P - v$, with $S = 24$, whether it arises as $(k, h) = (3, 0)$ or as $(k, h) = (2, 1)$. At order 8 the only such graph is $H_8$, with $S = 20$. At order 7 one has $h = 0$ in the equality case of $k = 1$, so the unique graph with $S = 16$ is $\Gamma_0$, while Cases 2 and 3 give at most 15 and 12. Since $8/3 < 3$ and $2 + 2/n < 3$ for $n \ge 11$, the bound $S \le 3n$ holds for every order $n \ge 2$, with equality only for $P$.

The case division does not use $n \ge 7$. At order 4 there is no cycle of length five or six, so only the chordal bound applies and $S \le 6$. Equality holds for a tree, and the only trees on four vertices

are $P_4$ and $K_{1,3}$. At order 6 the induced 6-cycle branch contributes only $C_6$, with $S = 12$. In the induced 5-cycle branch one has $h + j = 1$, hence $b = 0$ and $S = 12 - j - m \leq 12$, with equality precisely when $j = m = 0$ and $h = 1$, the graph $C_5$ with one leaf. The chordal branch contributes at most 10. Thus the maximum is 12, attained by exactly two graphs. □

*Remark* 3.7. Karthick and Maffray describe the structure of graphs with no induced $P_6$ and no induced 4-cycle, and deduce the colouring bound $\chi \leq \lceil 5\omega/4 \rceil$ [13]. Their Theorems 3.2 and 3.3 describe the neighbourhood, on an induced $C_5$ or $C_6$, of a vertex outside the cycle. Specialising the 6-cycle description to graphs with no 4-cycle subgraph excludes a consecutive triple, a consecutive quadruple and the full cycle as a support, and leaves an antipodal pair. That is the trace statement at the beginning of Case 1. The constraint on the second layer, the bound $h \leq 1$ when $k \in \{2, 3\}$, and the exact function $M(n)$ are further arguments. They are not consequences of the local support list, and the colouring theorem is not a substitute for the extremal statement under the subgraph prohibition.

*Remark* 3.8. Brandstädt and Hoàng analysed $(P_6, C_4)$-free graphs through clique cutsets [2]. In that language $C_4$ is an induced cycle. A graph in the class with no clique cutset has every induced $C_5$ dominating, and the presence of an induced $C_6$ that fails to dominate forces the graph to be the join of a blow-up of the Petersen graph and a clique, possibly empty. Theorem 3.6 is a different statement. The prohibition here is that of a 4-cycle as a subgraph, and the conclusion is the exact maximum of $S$ over every connected graph in the resulting class, including graphs that have clique cutsets. A nontrivial blow-up of the Petersen graph, and a join of the Petersen graph with a nonempty clique, each contain a 4-cycle subgraph, so neither construction enters the class.

*Remark* 3.9. An alternative proof of $S \leq 3n$, with equality only for the Petersen graph, fixes a vertex of eccentricity at most 2 by Lemma 3.5 and charges the cross edges in the second distance layer. The case division above already gives that inequality and the equality case, so the charging argument is not repeated.

## 4. Radius four

**Lemma 4.1.** *Let $G$ be connected, strong $C_4$-free and $P_8$-free. If an induced 8-cycle dominates $G$ and $r(G) \geq 4$, then $G = C_8$.*

*Proof.* Write $C = c_0c_1 \cdots c_7c_0$, with indices modulo 8. For an outside vertex $y$ the trace $A_y = N(y) \cap V(C)$ is nonempty.

The trace $A_y$ is not contained in any three consecutive vertices of $C$. Suppose $A_y \subseteq \{c_i, c_{i+1}, c_{i+2}\}$. The ends $c_i$ and $c_{i+2}$ are not both in $A_y$, since $y, c_i, c_{i+1}, c_{i+2}$ would be a 4-cycle. Thus $A_y$ is a singleton or a consecutive pair. A singleton $\{c_i\}$ gives the induced path

$$y, c_i, c_{i+1}, c_{i+2}, c_{i+3}, c_{i+4}, c_{i+5}, c_{i+6}.$$

A consecutive pair $\{c_i, c_{i+1}\}$ gives the induced path

$$y, c_i, c_{i-1}, c_{i-2}, c_{i-3}, c_{i-4}, c_{i-5}, c_{i-6},$$

in which the omitted contact $c_{i+1}$ does not lie on the path. Either path is an induced $P_8$.

Fix a cycle vertex, say $c_0$. The vertices at cycle distance greater than two from $c_0$ are the three consecutive vertices $c_3, c_4, c_5$. The trace of any outside vertex meets the complementary set $\{c_7, c_0, c_1, c_2, c_6\}$, so every outside vertex is within distance three of $c_0$. The same holds with $c_0$ replaced by any vertex of $C$.

Suppose $w$ lies outside $C$, and write $A = N(w) \cap V(C)$. No two vertices of $A$ are at cycle distance two, for they would form a 4-cycle with $w$. A set of vertices of $C$ in which every two distinct vertices are consecutive has size at most two, and every set of size at most two that is a singleton or a consecutive pair is contained in a triple of consecutive vertices. Both possibilities

have been excluded. Therefore $A$ contains two vertices at cycle distance three or four. Rotating, take them to be $c_0$ and $c_d$ with $d \in \{3, 4\}$.

The antipode of $c_0$ is $c_4$. If $d = 4$, the path $c_0 w c_4$ has length two. If $d = 3$, then $c_3 \sim c_4$, so $c_0 w c_3 c_4$ has length three. Every other vertex of $C$ is already at cycle distance at most three from $c_0$. Every outside vertex is within distance three of $c_0$. Thus the eccentricity of $c_0$ is at most three, contradicting $r(G) \geq 4$.

Hence $C$ has no vertex outside it, and $G = C_8$. □

*Remark* 4.2. The proof of Lemma 4.1 does not use a machine enumeration. An independent count, with the labelled cycle fixed, is the script `paper/scripts/c8_configurations.py`. A trace is admissible when the graph formed by the cycle and one vertex of that trace has no 4-cycle subgraph and no induced $P_8$. Configurations are built by adding admissible traces in increasing bitmask order, joining the new vertex to an arbitrary subset of the outside vertices already chosen, and discarding any graph that acquires a 4-cycle subgraph or an induced $P_8$. The script finds 32 admissible nonempty traces and 790 configurations, including the bare cycle. By order $8, \ldots, 13$ the counts are $1, 32, 262, 384, 107, 4$. It finds 73 isomorphism classes, and only the bare cycle has radius at least four. That count is not a hypothesis of Theorem 4.3.

**Theorem 4.3.** *Let $G$ be connected and strong $C_4$-free, with $r(G) = 4$ and $p(G) = 7$. Then either $G \cong C_8$, or $G$ is obtained from a 6-cycle $C = v_0 v_1 \cdots v_5$ by the following construction and by no further vertices or edges. For each $i$ there is a nonempty set $C_i$ of vertices outside $C$. Every vertex of $C_i$ is adjacent to $v_i$ and to no other vertex of $C$. There is no edge between $C_i$ and $C_j$ for $i \neq j$, and $G[C_i]$ is a matching together with isolated vertices. For each edge of $C$ there is at most one further vertex, adjacent precisely to the two ends of that edge.*

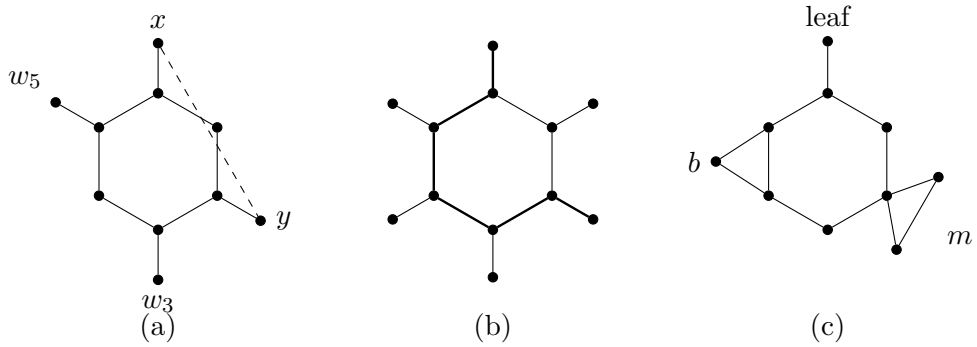



FIGURE 2. The radius-four case of Theorem 4.3. (a) If $x$ and $y$ have singleton traces $\{v_0\}$ and $\{v_2\}$ on the dominating 6-cycle, the dashed edge $xy$ completes the induced path $w_5, v_5, v_0, x, y, v_2, v_3, w_3$. The vertices $w_5$ and $w_3$ are the distance-four witnesses of the proof. (b) One leaf at each cycle vertex gives $S = 2n$. The heavy path has seven vertices. (c) On the same cycle: an isolated leaf, a matching edge inside one set $C_i$, and a vertex glued to a cycle edge. The leaf does not contribute to $b + m$. The matching edge and the glued vertex each increase $2n - S = b + m$ by 1.

*Proof.* Theorem 2.4 gives $p(G) \geq 7$, which agrees with the hypothesis. Apply Theorem 2.5 with $k = 8$. There is a connected dominating set $X$ such that $G[X]$ is $P_6$-free or $G[X] \cong C_8$.

If $G[X] \cong C_8$, Lemma 4.1 gives $G = C_8$.

If $G[X]$ is $P_6$-free, apply Lemma 3.5 inside $G[X]$. A vertex of eccentricity at most two in $G[X]$ has eccentricity at most three in $G$, because $X$ dominates $G$, contradicting $r(G) = 4$. Therefore

$G[X] = C_6$. Let $C = v_0 \cdots v_5$ be this dominating induced 6-cycle, indices modulo 6, and let $W = V(G) \setminus V(C)$.

*Traces.* As in the proof of Lemma 3.4, strong $C_4$-freeness allows only three kinds of trace $N(w) \cap C$: a single vertex, a consecutive pair $\{v_i, v_{i+1}\}$, or an antipodal pair. If some $w$ has trace $\{v_i, v_{i+3}\}$, then $v_i$ is within distance two of every vertex of $C$, hence within distance three of every vertex of $G$, a contradiction. So there are no antipodal traces. Let $C_i$ be the set of vertices with trace $\{v_i\}$.

*Witnesses.* Fix $i$. Since $r(G) = 4$, some vertex is at distance at least four from $v_{i+3}$. Every vertex of $C$ is within distance three of $v_{i+3}$, and a vertex $w \in W$ is at distance at most $1 + \min\{d_C(v_{i+3}, x) : x \in N(w) \cap C\}$ from $v_{i+3}$, which is at most three unless the trace of $w$ is $\{v_i\}$. So there is $w_i \in C_i$ at distance four from $v_{i+3}$. Every neighbour $z \in W$ of $w_i$ lies in $C_i$: if $z$ had a neighbour among $v_{i+2}, v_{i+3}, v_{i+4}$, then $d(w_i, v_{i+3}) \leq 3$; if $z$ had the neighbour $v_{i-1}$ or $v_{i+1}$, then $w_i, v_i, v_{i\pm1}, z$ would be a 4-cycle. So $N(w_i) \subseteq \{v_i\} \cup C_i$.

*Edges in $W$.* Figure 2 records the two configurations used below. Let $x, y \in W$ be adjacent.

- $x \in C_0$, $y \in C_{\pm1}$: $x, v_0, v_{\pm1}, y$ is a 4-cycle.
- $x \in C_0$, $y \in C_2$: $w_5, v_5, v_0, x, y, v_2, v_3, w_3$ is an induced $P_8$.
- $x \in C_0$, $y \in C_3$: $w_1, v_1, v_0, x, y, v_3, v_4, w_4$ is an induced $P_8$. By symmetry, vertices of $C_i$ and $C_j$ with $i \neq j$ are never adjacent.
- $x$ with trace $\{v_0, v_1\}$ and $y \in C_j$: for $j \in \{0, 1\}$ the vertices $v_0, v_1, x, y$ span a 4-cycle; for $j = 5$ (respectively $j = 2$) the cycle $x, v_0, v_5, y$ (respectively $x, v_1, v_2, y$) is a 4-cycle; for $j = 3$, $w_5, v_5, v_0, x, y, v_3, v_2, w_2$ is an induced $P_8$, and $j = 4$ is symmetric.
- $x, y$ with consecutive-pair traces $\{v_0, v_1\}$ and $\{v_j, v_{j+1}\}$: $j = 0$ gives two common neighbours; $j = \pm1$ and $j = \pm2$ give 4-cycles as above; $j = 3$ gives the induced $P_8$ $w_2, v_2, v_1, x, y, v_4, v_5, w_5$.

In each displayed path the non-edges follow from the traces and from $N(w_i) \subseteq \{v_i\} \cup C_i$. So every edge inside $W$ joins two vertices of the same class $C_i$, and these edges form a matching. A consecutive-pair vertex is adjacent to no vertex of any $C_j$ and to no other consecutive-pair vertex. Two vertices with the same consecutive-pair trace would have two common neighbours, so each cycle edge carries at most one such vertex, and that vertex is adjacent precisely to the two ends of the edge. Each class $C_i$ is nonempty. This is the stated construction.

Conversely, let $G$ be $C_8$, or let $G$ be built from $C = v_0 \cdots v_5$ as in the statement. The cycle $C_8$ has radius 4, induced-path order 7 and no 4-cycle subgraph. Now consider the construction on $C$. There is no edge between distinct sets $C_i$, and every vertex of $C_i$ meets $C$ only at $v_i$. A cycle that meets $C_i$ is therefore contained in $G[C_i \cup \{v_i\}]$. In that subgraph the edges are a star at $v_i$ together with a matching, so the only cycles are the triangles formed by $v_i$ and one matching edge.

Let $y$ be a vertex added on the edge $\{v_j, v_{j+1}\}$, so $N(y) = \{v_j, v_{j+1}\}$. Every cycle through $y$ uses both edges at $y$, and the rest of the cycle is a $v_j$–$v_{j+1}$ path avoiding $y$. A 4-cycle through $y$ is then a path of length two between $v_j$ and $v_{j+1}$ that avoids $y$: the two ends of the edge have a common neighbour other than $y$. No vertex of any $C_i$ is adjacent to two vertices of $C$, and the construction places at most one vertex on each cycle edge. Those two ends therefore have no common neighbour other than the vertex added on that edge. Hence no 4-cycle passes through $y$. The only cycle that uses no vertex outside $C$ is $C$ itself. Thus $G$ has no 4-cycle subgraph.

Let $P$ be an induced path. An isolated vertex of some $G[C_i]$ has degree one, so it is an end of $P$ if it lies on $P$. A vertex of a matching edge in $C_i$ has neighbourhood $\{v_i, x'\}$, a clique. It is therefore an end of $P$ whenever $P$ has a third vertex, and $P$ cannot contain both ends of the matching edge together with $v_i$. A vertex added on a cycle edge has neighbourhood equal to that edge, again a clique, so it too is an end of any induced path that contains it, and the path cannot contain both ends of the cycle edge. Consequently every induced path of order at least three

meets $V(G) \setminus V(C)$ in at most two vertices, both ends of the path. The remaining vertices lie on $C$, which has no induced path on six vertices. Hence $p(G) \leq 7$. Since each $C_i$ is nonempty, one may choose $x \in C_0$ and $y \in C_2$. The path $x, v_0, v_5, v_4, v_3, v_2, y$ is induced: $x$ and $y$ meet the path only at $v_0$ and $v_2$, and $C$ contributes no chord. Thus $p(G) = 7$.

There is no edge between distinct sets $C_i$, and a vertex added on a cycle edge has degree two on that edge, so it does not shorten distances along $C$. The distance from $v_i$ to $v_{i+3}$ is therefore three. A vertex of $C_{i+3}$ is at distance four from $v_i$, and the eccentricity of $v_i$ is at least four. It is at most four: every vertex of $C_j$ is at distance $d_C(v_i, v_j) + 1 \leq 4$ from $v_i$, and every vertex added on a cycle edge is within distance four of $v_i$. A vertex of $C_i$ is at distance five from a vertex of $C_{i+3}$. A vertex $y$ added on the edge $\{v_j, v_{j+1}\}$ has only those two neighbours. The distances from $v_j$ and from $v_{j+1}$ to $v_{j+3}$ are 3 and 2, neither neighbour of $y$ is adjacent to $v_{j+3}$, and $y, v_{j+1}, v_{j+2}, v_{j+3}$ is a path of length three, so a vertex of $C_{j+3}$ is at distance four from $y$. Every eccentricity is at least four, and $r(G) = 4$. □

**Corollary 4.4.** *Let $G$ be connected and strong $C_4$-free with $r(G) = 4$ and $p(G) = 7$. In the construction of Theorem 4.3, let $b$ be the number of vertices added on edges of the 6-cycle and let $m$ be the number of matching edges in the sets $C_i$. Then $S(G) = 2n - b - m$. In particular $S(G) \leq 2n$, so $\ell(G) \leq 2$, and* RAD(4) *holds.*

*Proof.* For $G \cong C_8$ one has $n = 8$, $e = 8$, $T = 0$ and $S = 16 = 2n$, which is the formula with $b = m = 0$. In the construction, write $a = \sum_i |C_i|$. Then $n = 6 + a + b$. The edges are the six cycle edges, $a$ edges from the sets $C_i$ to the cycle, $m$ matching edges and $2b$ edges incident with the vertices added on cycle edges, so $e = 6 + a + 2b + m$. Each matching edge forms one triangle with its cycle vertex, and each added vertex forms one triangle with its cycle edge, so $T = b + m$. Lemma 2.1(3) gives $S = 2e - 3T = 2n - b - m$. □

**Corollary 4.5.** *Let $G$ be connected and strong $C_4$-free with $r(G) = 4$ and $p(G) = 7$. Then $S(G) = 2n$ if and only if $G \cong C_8$, or $G$ is obtained from $C_6$ by attaching at least one leaf to each cycle vertex.*

*Proof.* By Corollary 4.4, $S = 2n$ if and only if $G \cong C_8$ or, in the construction, $b = m = 0$. In the latter case every vertex outside the cycle is a leaf, and each $C_i$ is nonempty, so each cycle vertex has at least one leaf. The converse is the case $b = m = 0$ of the construction, together with $C_8$, both of which have $S = 2n$ and satisfy the numerical hypotheses by Theorem 4.3. □

*Remark* 4.6. If $G$ is connected and strong $C_4$-free with $r(G) = 4$, $p(G) \geq 8$ and $\ell(G) \leq 4$, then $p(G) \geq 8 = r(G) + 4 \geq r(G) + \ell(G)$. This comparison is not a structural description of $G$.

## 5. Triangle-free graphs of induced-path order at most six

In a triangle-free graph, a 4-cycle subgraph is an induced 4-cycle, and every clique has order at most two. The next classification is Lemma 3.4 of Chen and Xu [5]. Their family $\mathfrak{F} = \{F_0, F_0', F_1, \ldots, F_{12}\}$ is defined by the constructions in the proofs of their Lemmas 3.1–3.3; $F_5$ is the Petersen graph, and the parameters $t$ and $t'$ attached to $F_2$ and $F_0'$ are positive integers. The edge counts below use those constructions rather than the accompanying figure.

**Lemma 5.1** (Chen and Xu [5, Lemma 3.4])**.** *Let $G$ be a connected $(P_7, C_4, C_3)$-free graph with no clique cutset. If $G$ is not complete, then $G \in \mathfrak{F}$.*

The members of $\mathfrak{F}$ admit the following descriptions, taken from those proofs.

- $F_1$ is $C_5$. The graph $F_2$ is obtained from a 5-cycle $v_1 \cdots v_5$ by adding stable sets $A_1$ and $A_3$ of equal positive order $t$, with a perfect matching between them, each vertex of $A_1$ adjacent only to $v_1$ and its match, and each vertex of $A_3$ adjacent only to $v_3$ and its match.

- On the same 5-cycle, $F_3$ has three further vertices inducing the path $a_3a_1a_4$, with $a_i$ adjacent to $v_i$. The graph $F_4$ has four further vertices inducing the path $a_5a_3a_1a_4$, with the same spokes. The graph $F_5$ is the Petersen graph.
- $F_0$ is $C_6$. The graph $F_0'$ is obtained from a 6-cycle $v_1 \cdots v_6$ by adding stable sets $Y_1$ and $Y_4$ of equal positive order $t'$, with a perfect matching between them, each vertex of $Y_1$ adjacent only to $v_1$ and its match, and each vertex of $Y_4$ adjacent only to $v_4$ and its match.
- $F_6$ is $C_7$. On a 7-cycle $v_1 \cdots v_7$, write $x_i$ for a vertex with neighbourhood $\{v_i, v_{i+3}\}$ on the cycle. Then $F_7$ is the cycle plus $x_1$; $F_8$ is the cycle plus nonadjacent $x_1, x_2$; $F_9$ is $F_8$ plus a vertex adjacent only to $x_1$ and $x_2$; $F_{10}$ is the cycle plus nonadjacent $x_1, x_5$; $F_{11}$ is the cycle plus a stable set $\{x_1, x_2, x_5\}$; and $F_{12}$ is $F_{11}$ plus a vertex adjacent only to $x_1$ and $x_2$.

Chen and Xu record that $F_0$, $F_1$, the graph $F_2$ with $t = 1$, $F_3$ and $F_4$ are induced subgraphs of the Petersen graph, and that $F_6, \ldots, F_{11}$ are induced subgraphs of $F_{12}$. The graphs $F_6, \ldots, F_{12}$ are the extensions of $C_7$ by vertices of cycle-distance three. The graphs $F_2$ and $F_0'$ are the matching extensions of one pair of opposite vertices on $C_5$ and on $C_6$.

**Lemma 5.2.** *Every graph in $\mathfrak{F}$ other than the Petersen graph satisfies $2e \le 3n-3$. The Petersen graph satisfies $2e = 3n$.*

*Proof.* The constructions give the following orders and sizes. For $F_2$ and $F_0'$ the parameters $t, t'$ are at least one.

| Graph | $n$ | $e$ | $2e$ | $3n-2e$ |
|---|---|---|---|---|
| $F_1 = C_5$ | 5 | 5 | 10 | 5 |
| $F_2$ | $5+2t$ | $5+3t$ | $10+6t$ | 5 |
| $F_3$ | 8 | 10 | 20 | 4 |
| $F_4$ | 9 | 12 | 24 | 3 |
| $F_5$ | 10 | 15 | 30 | 0 |
| $F_0 = C_6$ | 6 | 6 | 12 | 6 |
| $F_0'$ | $6+2t'$ | $6+3t'$ | $12+6t'$ | 6 |
| $F_6 = C_7$ | 7 | 7 | 14 | 7 |
| $F_7$ | 8 | 9 | 18 | 6 |
| $F_8$, $F_{10}$ | 9 | 11 | 22 | 5 |
| $F_9$, $F_{11}$ | 10 | 13 | 26 | 4 |
| $F_{12}$ | 11 | 15 | 30 | 3 |

The five edges of $F_3$ beyond $C_5$ are the three spokes and the two edges of $a_3a_1a_4$. The seven edges of $F_4$ beyond $C_5$ are the four spokes and the three edges of $a_5a_3a_1a_4$. Each vertex $x_i$ contributes two edges, and each vertex placed on a pair $x_ix_j$ contributes two further edges. The Petersen graph is the only row with difference zero. □

**Lemma 5.3.** *Let $F$ be connected and strong $C_4$-free, and let $s$ be a vertex at which no induced $P_4$ starts. Write $m = e(F[N(s)])$. Then $S(F) \le 2(|F|-1) - m$.*

*Proof.* A vertex at distance 3 from $s$ would end a geodesic, hence an induced $P_4$, starting at $s$. So $V = \{s\} \cup N_1 \cup N_2$ with $N_i$ the $i$-th distance layer. Each $y \in N_2$ has a unique parent $a \in N_1$. A neighbour $z \in N_2$ of $y$ with a different parent would give the induced path $sayz$. So $y$ has at most one neighbour in $N_2$, a sibling (two siblings would be common neighbours of $a$ and $y$), and $N(y)$ is a clique; thus $\lambda(y) = 1$. For $a \in N_1$ we have $N(a) \subseteq \{s, \bar{a}\} \cup C_a$, where $C_a$ is the set of children of $a$ and $\bar{a}$ is the neighbour of $a$ in $N_1$, if any. Since $s$ and $\bar{a}$ lie in one unit, $\lambda(a) \le 1 + |C_a|$. Finally $\lambda(s) = |N_1| - m$. Summing, $S \le (|N_1| - m) + (|N_1| + |N_2|) + |N_2| = 2(|F|-1) - m$. □

**Theorem 5.4.** *Let $G$ be a connected triangle-free strong $C_4$-free $P_7$-free graph. Then $2e(G) \leq 3|V(G)|$, with equality if and only if $G$ is the Petersen graph. In particular every connected triangle-free strong $C_4$-free graph of induced-path order at most 6 satisfies $\mathrm{GEN}_3$, and so does every connected bipartite strong $C_4$-free graph of induced-path order at most 6. A bipartite graph in the class has strict inequality.*

*Proof.* The proof is by induction on $n = |V(G)|$. The graph $K_2$ has $2e = 2 < 3n$. Suppose $n \geq 3$, and write $S = 2e$, which is legitimate because $G$ is triangle-free.

Suppose first that $v$ is a cut vertex. For each component $C$ of $G - v$ let $\rho_C$ be the largest order of an induced path in $G[C \cup \{v\}]$ that starts at $v$. Paths from two components form an induced path of $G$ on $\rho_C + \rho_{C'} - 1 \leq 6$ vertices, so some component $C_1$ has $\rho_{C_1} \leq 3$. Let $G_1 = G[C_1 \cup \{v\}]$ and $G_2 = G - C_1$, with orders $n_1$ and $n_2$. Then $n = n_1 + n_2 - 1$ and $n_1 \geq 2$. Lemma 2.3 gives $S(G) = S(G_1) + S(G_2)$. Lemma 5.3 gives $S(G_1) \leq 2(n_1 - 1)$. The graph $G_2$ is a smaller connected triangle-free strong $C_4$-free $P_7$-free graph, so the inductive hypothesis gives $S(G_2) \leq 3n_2$. Therefore

$$S(G) \leq 2(n_1 - 1) + 3n_2 = 3n - (n_1 - 1) < 3n.$$

If $G$ has a cut vertex, this estimate gives the theorem, and equality does not hold. Assume for the remainder of the argument that $G$ has no cut vertex.

Suppose that $\{u, v\}$ is a clique cutset of order two. Then $uv$ is an edge. Since neither vertex is a cut vertex, every component of $G - \{u, v\}$ is adjacent to both $u$ and $v$, and there are at least two such components. A shortest $u$–$v$ path through one component has at least four edges: one internal vertex would form a triangle with $uv$, and two internal vertices would form a 4-cycle. In two components, choose shortest paths, and on each path let $x, y, z$ be the three vertices immediately preceding $v$, in that order from the $u$-side. The same choice in the second component produces $x', y', z'$. The path $x, y, z, v, z', y', x'$ is an induced $P_7$. Indeed, each side is a subpath of a shortest path, so it has no chord, and there is no edge between $\{x, y, z\}$ and $\{x', y', z'\}$ because those sets lie in different components of $G - \{u, v\}$. This contradicts $p(G) \leq 6$.

Thus $G$ has no clique cutset. A triangle-free graph has no 4-cycle subgraph if and only if it has no induced 4-cycle, so Lemma 5.1 applies: $G$ is complete or belongs to $\mathfrak{F}$. The only complete triangle-free graph on at least two vertices is $K_2$, already strict. Lemma 5.2 gives $2e \leq 3n$ for every graph in $\mathfrak{F}$, with equality only for the Petersen graph.

A bipartite graph is triangle-free. The Petersen graph is not bipartite, so the inequality is strict for every connected bipartite strong $C_4$-free $P_7$-free graph.

Equality holds only for the Petersen graph. In particular, a graph with a clique cutset lies strictly below the bound, whether the cutset is a single vertex or an edge. □

**Theorem 5.5.** *Let $f(n)$ be the maximum of $2e(G)$ over connected triangle-free strong $C_4$-free $P_7$-free graphs of order $n$. For every $n \geq 10$,*

$$f(n) = 3n - \min\{n - 10,\ 5 + \mathbf{1}_{\{n \text{ even}\}}\}.$$

*Proof.* Write $\eta(H) = 3|V(H)| - 2e(H)$ for a triangle-free graph $H$. The argument is the cut-vertex analysis of Theorem 5.4, continued until the core is one graph from the table of Lemma 5.2.

*Reduction to one core.* Let $G$ be connected, triangle-free, strong $C_4$-free and $P_7$-free, and let $s$ be a cut vertex. As in Theorem 5.4, path concatenation supplies a component $C$ of $G - s$ such that every induced path of $G[C \cup \{s\}]$ that starts at $s$ has order at most three. The first distance layer of this side is independent, since an edge there would form a triangle with $s$. There is no vertex at distance three or more from $s$, for such a vertex would end an induced $P_4$ at $s$. A vertex of the second layer has a unique parent: a second parent, together with $s$, would span a 4-cycle. It has no neighbour in its layer. An edge joining vertices of distinct parents is the induced path of order four recorded in the proof of Lemma 5.3, and an edge joining two children

of the same parent forms a triangle with that parent. Thus $G[C \cup \{s\}]$ is a tree. Deleting the $k = |C|$ vertices of the side deletes exactly $k$ edges. Iterate. The remaining connected graph $Q$ has no cut vertex. The edge-separator argument of Theorem 5.4 excludes a clique cutset of order two in $Q$, and $Q$ is triangle-free, so it has no clique cutset at all. Lemma 5.1 therefore places $Q$ among the complete graphs and the graphs of Lemma 5.2. The only complete triangle-free graphs are $K_1$ and $K_2$. Reversing the deletions attaches trees at single vertices. With $h = |V(Q)|$ and $e(G) = e(Q) + (n - h)$,

$$3n - 2e(G) = \eta(Q) + n - h.$$

*Upper bound.* Every fixed row of the table in Lemma 5.2 satisfies $\eta \geq h - 10$. Each such graph has order at most 10 except $F_{12}$, which has order 11 and $\eta = 3$, and $3 \geq 11 - 10$. The same inequality holds for $K_1$ and $K_2$, with deficits 3 and 4. For every such core, $\eta(Q) + n - h \geq n - 10$. The two parameterised rows are $F_2$, of odd order and deficit 5, and $F_0'$, of even order and deficit 6. If the core is $F_2$ and the final order is even, then at least one vertex has been attached, so the displayed deficit is at least 6. If the core is $F_2$ and the final order is odd, the deficit is at least 5. If the core is $F_0'$, the deficit is at least 6. In every case

$$3n - 2e(G) \geq \min\bigl\{n - 10,\ 5 + \mathbf{1}_{\{n \text{ even}\}}\bigr\}.$$

*Attainment.* Attach $n - 10$ leaves to a single vertex of the Petersen graph. The resulting graph is connected and triangle-free, and a leaf lies on no cycle, so it remains strong $C_4$-free. An induced path that uses one leaf has the leaf at an end and the rest in the Petersen graph, which is $P_6$-free, so the path has at most six vertices. An induced path that uses two leaves has three vertices, because both leaves meet only the chosen vertex. The deficit is $n - 10$. For the other deficit, use $F_2$ at odd orders and $F_0'$ at even orders. In the descriptions preceding Lemma 5.2, each of these graphs consists of internally disjoint three-edge paths between two hubs, together with one additional two-edge path in the case of $F_2$. Every cycle uses two of those paths and therefore has length at least five. An induced path that contains at most one hub meets that hub in at most two of the paths and so has at most five vertices. An induced path that contains both hubs uses at most one path between them, a segment of at most three edges, and each end extends by at most one further edge before the next vertex meets the opposite hub and creates a chord. Its order is at most six. The table of Lemma 5.2 gives deficits 5 and 6 respectively. The smaller of the two deficits is therefore attained, which is $f(n)$. □

**Theorem 5.6.** *Let $G$ be a connected strong $C_4$-free $P_7$-free graph with at most one triangle. Then $S(G) \leq 3n$. In particular the same bound holds for a graph with exactly one triangle.*

*Proof.* Suppose $G$ is a counterexample of minimum order in the stated class, so $S(G) \geq 3n + 1$. Theorem 5.4 forces $G$ to contain a triangle, hence exactly one. The short-side argument at a cut vertex in Theorem A.1 applies inside this hereditary class. Let $v$ be a cut vertex. For each component $C$ of $G - v$ let $\rho_C$ be the largest order of an induced path in $G[C \cup \{v\}]$ with end $v$. Two such paths from different components form an induced path on $\rho_C + \rho_{C'} - 1 \leq 6$ vertices, so some component $C_1$ has $\rho_{C_1} \leq 3$. Put $G_1 = G[C_1 \cup \{v\}]$ and $G_2 = G - C_1$, with $n_i = |G_i|$, so $n = n_1 + n_2 - 1$ and $n_1 \geq 2$. Both graphs are connected, smaller than $G$, strong $C_4$-free and $P_7$-free, and each has at most one triangle. Lemma 5.3 gives $S(G_1) \leq 2(n_1 - 1)$, and minimality gives $S(G_2) \leq 3n_2$. By Lemma 2.3,

$$S(G) \leq 2(n_1 - 1) + 3n_2 = 3n - (n_1 - 1) < 3n.$$

Thus $G$ is 2-connected.

Let $v$ be a vertex of minimum degree among the three vertices of the triangle, write $d = d(v)$, and let $u$ and $w$ be the other two. The graph $H = G - v$ is connected and triangle-free. Apply the reduction in the proof of Theorem 5.5 to $H$, and write $Q$ for its core, $h = |V(Q)|$ and $\eta = \eta(Q)$.

Set

$$a = \big|N_G(v) \cap V(Q)\big|, \qquad c = \big|V(H) \setminus \big(V(Q) \cup N_G(v)\big)\big|,$$

and $t = |V(H)| - h$. The vertices of $H$ outside $Q$ are the $d-a$ neighbours of $v$ that miss $Q$, together with the $c$ vertices counted above, so $t = d - a + c$. The same reduction gives $e(H) = e(Q) + t$. Since $G$ has one triangle, $S(G) = 2e(G) - 3$ and $e(G) = e(H) + d$, and therefore

$$S(G) - 3n = 2d - 6 - \eta - t = d + a - 6 - \eta - c. \tag{2}$$

The core satisfies $a \le \eta + 3$ and, for every edge $xy$ of $Q$,

$$a + \min\{d_Q(x), d_Q(y)\} \le \eta + 5. \tag{3}$$

These bounds are read from the list in Lemma 5.2, using the descriptions that precede that lemma. If $Q$ is the Petersen graph, then $\eta = 0$. Every two nonadjacent vertices of the Petersen graph have a common neighbour, so two such vertices cannot both be adjacent to $v$. Thus $N(v) \cap V(Q)$ is a clique, necessarily of size at most two, and every degree in $Q$ is three. Both bounds follow. For each remaining fixed graph of order at least three one has $h \le 11$ and minimum degree at least two. The open neighbourhoods in $Q$ of any two vertices of $N(v) \cap V(Q)$ are disjoint: a common neighbour would form a 4-cycle with $v$. Hence $a \le \lfloor h/2 \rfloor \le 5$. Every degree in these graphs is at most four. The only rows with $\eta < 4$ are $F_4$ and $F_{12}$, both with $\eta = 3$. The path-and-spoke description of $F_4$ has maximum degree three. In $F_{12}$ the only vertices of degree four are $v_1$ and $v_5$ on the 7-cycle. Each is adjacent to two cycle vertices and to two of $\{x_1, x_2, x_5\}$, and $v_1$ is not adjacent to $v_5$, so every edge of $F_{12}$ has an end of degree at most three. This gives (3) and the bound on $a$. In each parameterised family every non-hub vertex is adjacent to a hub, and each hub has at most one neighbour in common with $v$, so $a \le 4$ even after both hubs are counted. Every edge has an end of degree two, because the edges are the internal edges of the paths between the hubs described above, and $\eta$ is 5 or 6. The graphs $K_1$ and $K_2$ satisfy both bounds directly: $K_1$ has $\eta = 3$ and no edge, and $K_2$ has $\eta = 4$.

If $u$ or $w$ lies outside $Q$, say $u$, the tree attachments give $u$ at most one neighbour in $Q$. Apart from $w$, no neighbour of $u$ in $H$ is adjacent to $v$, because the triangle is unique. Thus $d \le d(u) \le c + 3$. Substituting in (2) gives $S(G) - 3n \le a - \eta - 3$, and $a \le \eta + 3$ yields $S(G) - 3n \le 0$. If instead both $u$ and $w$ lie in $Q$, then $uw$ is an edge of $Q$, and every neighbour of either vertex that lies in $H - V(Q)$ lies in the set counted by $c$. Since $v$ has minimum degree among the three triangle vertices,

$$d \le 1 + c + \min\{d_Q(u), d_Q(w)\}.$$

Substituting in (2) and applying (3) to the edge $uw$ again yields $S(G) \le 3n$. □

**Corollary 5.7.** *Let $G$ be connected and strong $C_4$-free with $r(G) = 3$, $p(G) = 6$ and $\ell(G) > 3$. Then $G$ contains a triangle.*

*Proof.* If $G$ were triangle-free, Theorem 5.4 would give $S(G) \le 3n$. □

**Lemma 5.8.** *Let $G$ be a graph and let $K$ be a clique cutset of order $t$, with $G - K$ having $k \ge 2$ components. For each component $C$ let $G_C$ be the subgraph induced by $C \cup K$, and write $\Delta(H) = 2|V(H)| - 3 - e(H)$. Then*

$$\Delta(G) = \sum_C \Delta(G_C) + (k-1)\Big[\binom{t}{2} + 3 - 2t\Big].$$

*The bracket equals 1 if $t = 1$ and equals 0 if $t = 2$ or $t = 3$.*

*Proof.* The edges inside $K$ are counted $k$ times among the graphs $G_C$, and $|V(G)| = \sum_C |V(G_C)| - (k-1)t$, so

$$e(G) = \sum_C e(G_C) - (k-1)\binom{t}{2}.$$

Therefore

$$\begin{aligned}\Delta(G) &= 2\sum_C |V(G_C)| - 2(k-1)t - 3 - \sum_C e(G_C) + (k-1)\binom{t}{2} \\ &= \sum_C \big(\Delta(G_C) + 3\big) - 3 - 2(k-1)t + (k-1)\binom{t}{2} \\ &= \sum_C \Delta(G_C) + (k-1)\Big[\binom{t}{2} + 3 - 2t\Big].\end{aligned}$$

The bracket is $0 + 3 - 2 = 1$ for $t = 1$, $1 + 3 - 4 = 0$ for $t = 2$, and $3 + 3 - 6 = 0$ for $t = 3$. □

**Lemma 5.9.** *Let $G$ be a connected strong $C_4$-free $P_7$-free graph on $n \geq 2$ vertices, and write $\Delta(G) = 2n - 3 - e(G)$. Then $\Delta(G) \geq 0$. If $n \geq 4$, then $\Delta(G) \geq 1$, so $e(G) \leq 2n - 4$ and $S(G) \leq 4n - 8 - 3T(G)$. In particular $\ell(G) < 4$.*

*Proof.* A copy of $K_4$ contains a 4-cycle, so $\omega(G) \leq 3$. A diamond contains a 4-cycle, so every graph in the class is $(P_7, C_4, \text{diamond})$-free in the sense of Chen, Wu and Xu [6]. The proof is by induction on $n$. For $n = 2$ the graph is $K_2$ and $\Delta = 0$. For $n = 3$ the maximum of $e$ is 3, attained by $K_3$, and $\Delta \geq 0$.

Let $n = 4$. A graph on four vertices with five edges is $K_4$ minus an edge. Labelling the missing edge $cd$, the cycle $a\,c\,b\,d$ uses four edges of the graph. Thus $e \leq 4$ and $\Delta = 5 - e \geq 1$.

Now let $n \geq 5$, and assume the two claims for every smaller graph in the class. Every block $G_C$ arising from a clique cutset is itself connected, strong $C_4$-free and $P_7$-free, of order strictly less than $n$.

Suppose first that $v$ is a cut vertex, and that $G - v$ has $k \geq 2$ components. Lemma 5.8 with $t = 1$ gives $\Delta(G) = \sum_i \Delta(G_i) + (k-1)$. Each side has order at least 2, so $\Delta(G_i) \geq 0$ by induction, and $\Delta(G) \geq k - 1 \geq 1$.

Suppose next that a triangle $K$ is a clique cutset and $G - K$ has $k \geq 2$ components. Each side $G_i$ contains $K$ and at least one further vertex, so its order is at least 4. Lemma 5.8 gives $\Delta(G) = \sum_i \Delta(G_i)$, and induction supplies $\Delta(G_i) \geq 1$. Hence $\Delta(G) \geq k \geq 2$.

Suppose that $\{u, v\}$ is a clique cutset and $G - \{u, v\}$ has $k \geq 2$ components. Lemma 5.8 gives $\Delta(G) = \sum_i \Delta(G_i)$. Induction gives $\Delta(G_i) \geq 0$ for every side, and each side has order at least 3. If every side had deficit 0, then each side would have order 3 and exactly three edges, so each side would be a triangle. Two triangles that share the edge $uv$ span a 4-cycle on their two opposite vertices, which contradicts the hypothesis that $G$ is strong $C_4$-free. Some side therefore has positive deficit, and $\Delta(G) \geq 1$.

Suppose finally that $G$ has no clique cutset. Theorem 1.1 of [6] states that a connected $(P_7, C_4, \text{diamond})$-free graph with no clique cutset is the Petersen graph, or the graph $F$ defined below, or else has minimum degree at most $\max\{2, \omega - 1\}$. Here $\omega \leq 3$, so the last quantity is at most 2. The Petersen graph has $n = 10$ and $e = 15$, hence $\Delta = 2$. The graph $F$ is obtained from a 7-cycle $v_1 \cdots v_7$ by adding an independent set $\{y_1, y_2, y_3\}$ with

$$N(y_1) = \{v_1, v_4, v_5\}, \qquad N(y_2) = \{v_2, v_5, v_6\}, \qquad N(y_3) = \{v_3, v_6, v_7\}.$$

Thus $F$ has $n = 10$ and $e = 16$, hence $\Delta = 1$. The three triangles of $F$ are $y_1v_4v_5$, $y_2v_5v_6$ and $y_3v_6v_7$. If instead some vertex $v$ has degree at most 2, then $v$ is not a cut vertex, so $G - v$ is

connected and lies in the class. Deleting $v$ changes the deficit by

$$\Delta(G) - \Delta(G - v) = 2 - d(v) \geq 0.$$

For $n \geq 5$ the order of $G - v$ is at least 4, so induction gives $\Delta(G - v) \geq 1$ and therefore $\Delta(G) \geq 1$.

Lemma 2.1(3) gives $S = 2e - 3T$. For $n \geq 4$ one has $e \leq 2n - 4$, so $S \leq 4n - 8 - 3T \leq 4n - 8$ and $\ell \leq 4 - 8/n < 4$. For $2 \leq n \leq 3$ one has $e \leq 2n - 3$ and $S \leq 4n - 6$, so again $\ell < 4$. □

**Lemma 5.10.** *Let $G$ be a connected strong $C_4$-free $P_7$-free graph.*

1. *If $n \geq 5$ and $T(G) \leq 1$, then $\Delta(G) \geq 2$, so $e(G) \leq 2n - 5$. If $T(G) = 1$, then $S(G) \leq 4n - 13$.*
2. *If $n \geq 4$ and $T(G) \geq 2$, then $S(G) \leq 4n - 14$.*
3. *If $n \geq 5$ and $T(G) \geq 1$, then $S(G) \leq 4n - 13$.*
4. *If $n \geq 6$ and $T(G) \leq 2$, then $\Delta(G) \geq 2$.*

*Proof.* If $T \geq 2$ and $n \geq 4$, Lemma 5.9 gives $e \leq 2n-4$, hence $S = 2e-3T \leq 2(2n-4)-6 = 4n-14$. This is (2). Once (1) is known, $T = 1$ gives $S = 2e - 3 \leq 4n - 13$, and (2) is at least as strong, so (3) follows. It remains to prove $\Delta \geq 2$ when $n \geq 5$ and $T \leq 1$.

The proof is by induction on $n$. Let $n = 5$. A chordless cycle in a strong $C_4$-free graph has length at least 5, so the only possible hole on five vertices is a 5-cycle. A chord of $C_5$ produces a 4-cycle, and there is no room for a further vertex, so $G \cong C_5$. Then $e = 5$, $T = 0$ and $\Delta = 2$. If $G$ has no hole, then $G$ is chordal. A simplicial vertex of a connected chordal graph on at least two vertices is not a cut vertex, and its neighbourhood is a clique. That clique has order at most two: a simplicial vertex of degree three or more, together with three neighbours, contains a $K_4$ and therefore a 4-cycle. Deleting a simplicial vertex of degree one removes one edge and no triangle. Deleting a simplicial vertex of degree two removes two edges and exactly one triangle. Repeating until a single vertex remains gives $e = n - 1 + T$. Here $T \leq 1$, so $e \leq 5$ and $\Delta = 7 - e \geq 2$.

Now let $n \geq 6$, with $T(G) \leq 1$, and assume the deficit claim for every smaller graph in the class that has at most one triangle. By Lemma 5.9 every graph in the class of order at least 2 has nonnegative deficit, and every graph of order at least 4 has deficit at least 1.

Suppose $v$ is a cut vertex and $G - v$ has $k \geq 2$ components. Lemma 5.8 gives $\Delta(G) = \sum_i \Delta(G_i) + (k - 1)$. If this quantity were at most 1, then $k = 2$ and both sides would have deficit 0. A side of deficit 0 has order at most 3, so $n \leq 3 + 3 - 1 = 5$, a contradiction. Thus $\Delta(G) \geq 2$.

Suppose a triangle is a clique cutset, with $k \geq 2$ sides. Each side has order at least 4, so each has deficit at least 1 by Lemma 5.9, and Lemma 5.8 gives $\Delta(G) \geq 2$.

Suppose $\{u, v\}$ is a clique cutset and $G - \{u, v\}$ has $k \geq 2$ components, and suppose toward a contradiction that $\Delta(G) \leq 1$. Lemma 5.8 gives $\Delta(G) = \sum_i \Delta(G_i)$. A side of order at least 4 has deficit at least 1. A side of deficit 0 therefore has order 3 and three edges, so it is a triangle. Two such sides would give two triangles and a 4-cycle, which is impossible under $T(G) \leq 1$ and the strong $C_4$-free hypothesis. If every side had deficit at least 1, the sum would be at least $k \geq 2$. Hence exactly one side is a triangle, of deficit 0, and exactly one other side is present: a third side would contribute deficit at least 1 and would push the sum to at least 2. The hypothesis $T(G) \leq 1$ forces the other side $G_2$ to be triangle-free. Its order is $n - 1 \geq 5$. The inductive hypothesis gives $\Delta(G_2) \geq 2$, so $\Delta(G) \geq 2$, a contradiction. Thus $\Delta(G) \geq 2$.

Suppose finally that $G$ has no clique cutset. The Petersen graph has $T = 0$ and $\Delta = 2$. The exceptional graph $F$ of the previous proof has three triangles, so it does not fall under $T \leq 1$. In every remaining case Chen, Wu and Xu supply a vertex $v$ of degree at most 2. This vertex is not a cut vertex. It lies in at most one triangle, and $T(G - v) = T(G) - t(v) \leq 1$. The graph $G - v$ is connected, strong $C_4$-free and $P_7$-free, of order at least 5. Induction gives $\Delta(G - v) \geq 2$. Since $\Delta(G) = \Delta(G - v) + 2 - d(v) \geq \Delta(G - v)$, one has $\Delta(G) \geq 2$.

It remains to prove (4). Part (1) already gives the claim when $T \leq 1$. Suppose $T = 2$, and proceed by induction on $n$ from order six. Lemma 5.9 supplies $\Delta \geq 0$ for every connected graph in the class of order at least two, and $\Delta \geq 1$ for every such graph of order at least four.

Suppose $v$ is a cut vertex and $G - v$ has $k \geq 2$ components. Lemma 5.8 gives $\Delta(G) = \sum_i \Delta(G_i) + (k-1)$. A value at most 1 would require $k = 2$ and both sides of deficit 0, hence both of order at most three, and then $n \leq 5$. Thus $\Delta(G) \geq 2$. Suppose a triangle is a clique cutset. Each side has order at least four, the side deficits add, and Lemma 5.9 gives $\Delta(G) \geq 2$. Suppose $\{u, v\}$ is a clique cutset and $G - \{u, v\}$ has $k \geq 2$ components. The side deficits again add. A side of deficit 0 has order three and three edges, so it is a triangle. Two such sides would contain a 4-cycle, so there is at most one. If $\Delta(G) \leq 1$, there are exactly two sides, one a triangle and the other a graph of order $n - 1 \geq 5$ with at most one triangle. Part (1) gives that side deficit at least two, a contradiction. This handles every clique cutset.

Suppose $G$ has no clique cutset. Neither the Petersen graph nor the exceptional graph $F$ of Lemma 5.9 has exactly two triangles. Theorem 1.1 of Chen, Wu and Xu, in the form already used for Lemma 5.9, therefore supplies a vertex $x$ of degree at most two. The graph $J = G - x$ is connected, and $\Delta(G) = \Delta(J) + 2 - d(x)$. For $n \geq 7$ the inductive hypothesis gives $\Delta(J) \geq 2$ when $T(J) \leq 2$, while $T(J) \leq 1$ is the case of part (1); in both subcases $\Delta(G) \geq 2$. For $n = 6$ the same comparison leaves only the case $d(x) = 2$, $|V(J)| = 5$ and $\Delta(J) = 1$. Then $J$ has six edges. A hole in $J$ would be a 5-cycle, and any chord of that cycle is a 4-cycle, so $J$ is chordal. The simplicial deletion in the order-five argument above gives $e(J) = |V(J)| - 1 + T(J)$, hence $T(J) = 2$. Two triangles on five vertices cannot share an edge, for that graph contains a 4-cycle, and they cannot be disjoint, so they share exactly one vertex. Every pair of vertices of this graph has a common neighbour. Joining $x$ to any two of them creates a 4-cycle, which contradicts $d(x) = 2$. □

**Corollary 5.11.** *Let $G$ be a connected strong $C_4$-free graph with $\ell(G) \leq 4$ that violates $p(G) \geq r(G) + \ell(G)$. Then*

$$r(G) = 3, \qquad p(G) = 6, \qquad T(G) \geq 2, \qquad n \geq 17, \qquad 3 < \ell(G) \leq 4 - \frac{16}{n}.$$

*Whether any graph meets these conditions remains open.*

*Proof.* Write $r = r(G)$ and $p = p(G)$. Theorem 5.13 gives the inequality whenever $\ell \leq 3$ and whenever $r \leq 2$ and $\ell \leq 4$, so a counterexample has $\ell > 3$ and $r \geq 3$. A violation means $p \leq r + \lceil \ell \rceil - 1$. Since $\ell \leq 4$ one has $\lceil \ell \rceil \leq 4$, so $p \leq r + 3$. Theorem 2.4 gives $p \geq 2r - 1$, hence $r \leq 4$. If $r = 4$, then $p = 7$, and Corollary 4.4 gives $\ell \leq 2$. Thus $r = 3$ and $p \leq 6$. Theorem 2.4 gives $p \geq 5$. If $p \leq 5$, Theorem 3.6 gives $\ell \leq 3$. Hence $p = 6$, and $S > 3n$.

The graph is strong $C_4$-free and $P_7$-free. Theorem 5.4 and Theorem 5.6 force $T \geq 2$. If $n \leq 5$, part (2) of Lemma 5.10 and the bound $S \leq 4n - 6$ of Lemma 5.9 give $S \leq 3n$. If $T = 2$ and $n \geq 6$, part (4) of Lemma 5.10 gives $\Delta \geq 2$, so $e \leq 2n - 5$ and $S = 2e - 6 \leq 4n - 16$. If $T \geq 3$ and $n \geq 4$, Lemma 5.9 gives $S \leq 4n - 8 - 3T \leq 4n - 17$. In the remaining range $S \leq 4n - 16$, so $\ell \leq 4 - 16/n$. Combined with $S > 3n$ this forces $n \geq 17$. For every such order, $4 - 16/n < 4$, so $\ell \neq 4$. □

*Remark* 5.12. The degree bound in Theorem 1.1 of Chen, Wu and Xu does not by itself yield $S \leq 3n$. Lemmas 5.9 and 5.10 are the deficit estimates that follow from that theorem once clique cutsets of order 1, 2 and 3 are restored and the two exceptional graphs are counted. The resulting bound is $\ell < 4$. Part (4) of Lemma 5.10 is the strengthening that includes graphs with exactly two triangles. It is not the bound $S \leq 3n$ for every number of triangles. The case of $\mathrm{GEN}_3$ in which two or more triangles are present remains open. The supplement records a conditional reduction for graphs that contain a triangle.

**Theorem 5.13** (Theorem A)**.** *Let $G$ be connected and strong $C_4$-free with $n \geq 2$. Then $p(G) \geq r(G) + \ell(G)$ whenever $\ell(G) \leq 3$, whenever $r(G) \leq 2$ and $\ell(G) \leq 4$, whenever $r(G) = 4$ and $\ell(G) \leq 4$, and whenever $r(G) \geq 5$ and $\ell(G) \leq 4$. If $r(G) = 4$ and $p(G) = 7$, then $\ell(G) \leq 2$.*

*Proof.* Write $r = r(G)$ and $c = \lceil \ell(G) \rceil$. The last sentence is Corollary 4.4. For the inequality we need $p \geq r + c$ in the listed ranges.

- $r = 1$: Lemma 3.1.
- $r = 2$, $c \leq 2$: Lemma 3.2.
- $r = 2$, $c = 3$: if $p \leq 4$, then Lemma 3.3 gives $\ell \leq 2$, a contradiction. So $p \geq 5$.
- $r = 2$, $c = 4$: if $p \leq 5$, then Theorem 3.6 gives $\ell \leq 3$, a contradiction. So $p \geq 6$.
- $r = 3$, $c \leq 2$: Theorem 2.4 gives $p \geq 5$.
- $r = 3$, $c = 3$: if $p \leq 5$, then by Lemma 3.5 and $r = 3$ we have $G = C_6$, with $\ell = 2$, a contradiction. So $p \geq 6$. Theorem 3.6 gives $\ell \leq 3$ for every $P_6$-free graph in the class, which does not rule out $c = 3$.
- $r \geq 4$, $c \leq 3$: Theorem 2.4 gives $p \geq 2r - 1 \geq r + 3$.
- $r = 4$ and $\ell \leq 4$: Theorem 2.4 gives $p \geq 7$. If $p = 7$, Corollary 4.4 gives $\ell \leq 2$, so $p \geq r + \ell$. If $p \geq 8$, Remark 4.6 gives the same inequality.
- $r \geq 5$ and $\ell \leq 4$: Theorem 2.4 gives $p \geq 2r - 1 \geq r + 4 \geq r + c$.

□

Equality in REAL-133 occurs for $C_5$ ($r = 2$, $\ell = 2$, $p = 4$), the Petersen graph ($r = 2$, $\ell = 3$, $p = 5$) and $C_6$ ($r = 3$, $\ell = 2$, $p = 5$). We do not characterise all equality cases.

## 6. The reduction and the residual cases

For integers $K \geq 3$ and $r \geq 4$ consider the following statements.

GEN$_K$: every connected strong $C_4$-free graph $G$ with $p(G) \leq 2K$ satisfies $S(G) \leq Kn$.

RAD($r$): every connected strong $C_4$-free graph with $r(G) = r$ and $p(G) = 2r - 1$ satisfies $\ell(G) \leq r - 1$.

Corollary 4.4 is the case $r = 4$ of RAD($r$), in the stronger form $\ell \leq 2$.

**Corollary 6.1.** *Let $G$ be a connected strong $C_4$-free graph that violates Conjecture 1.1. Then $\ell(G) > 3$. Put $K = \lceil \ell(G) \rceil - 1 \geq 3$. Then either*

(a) *$r(G) \leq K$ and $p(G) \leq r(G) + K \leq 2K$, so $G$ violates* GEN$_K$*; or*
(b) *$r(G) = K + 1 \geq 4$ and $p(G) = 2r(G) - 1$, and $\ell(G) > r(G) - 1$, so $G$ violates* RAD$(K + 1)$.

*Proof.* Write $r = r(G)$, $p = p(G)$ and $c = \lceil \ell(G) \rceil = K + 1$. Then $\ell > 3$ by Theorem 5.13. A violation means $p \leq r + c - 1 = r + K$. Theorem 2.4 gives $2r - 1 \leq p \leq r + K$, so $r \leq K + 1$. If $r \leq K$, then $p \leq r + K \leq 2K$, and $S > Kn$ because $\ell > K$. If $r = K + 1$, then $2r - 1 \leq p \leq 2K + 1 = 2r - 1$, and $\ell > K = r - 1$. □

Since GEN$_K$ ignores the radius, it is stronger than alternative (a). The statement GEN$_K$ might fail for large $K$ even if REAL-133 holds; see Remark 2.5.

**Proposition 6.2.** *For $k \geq 3$ let $\theta_k$ consist of two vertices $x, y$ joined by $k$ internally disjoint paths $xa_ib_iy$. Then $\theta_k$ is strong $C_4$-free, $n = 2k + 2$, $r = 3$, $p = 6$ and $S = 6k = 3n - 6$. Thus $p - r - \ell = 6/n \to 0$.*

*Proof.* Every cycle consists of two complete branches and so has length 6; in particular there is no 4-cycle. There are no triangles, so $S = 2e = 6k$. Every vertex has eccentricity 3: $d(x, y) = 3$ and $d(a_i, b_j) = 3$ for $i \neq j$. The path $a_jxa_ib_iyb_l$ (with $j, i, l$ distinct) is induced. Conversely, let $P$ be an induced path. If $P$ avoids $x$ and $y$, it lies inside one edge $a_ib_i$. If $P$ contains $x$ but not $y$, then $x$ has at most two neighbours on $P$, each followed on $P$ by at most its $b$-vertex, so $|P| \leq 5$.

If $P$ contains $x$ and $y$, then every $a_j$ on $P$ is adjacent on $P$ to $x$ and every $b_j$ to $y$, at most one branch lies entirely on $P$, and so $P$ consists of $x$, $y$, at most one full branch and at most one further neighbour of each of $x$ and $y$. Hence $|P| \le 6$. □

Thus no argument with a uniform positive margin in $\ell$ can prove $\mathrm{GEN}_3$, already among graphs of radius 3. The resulting division of REAL-133 is recorded once, in Table 1.

Let $3 < \ell \le 4$, and suppose $G$ violates REAL-133. Corollary 5.11 places every such counterexample in the cell $r = 3$, $p = 6$, $T \ge 2$, $n \ge 17$ and $3 < \ell \le 4 - 16/n$. In particular $\ell = 4$ does not occur. The parts of that argument that are not the Erdős–Saks–Sós bound are Theorems 3.6, 4.3, 5.4, 5.5 and 5.6, together with Lemmas 5.9 and 5.10. The inequality $\mathrm{GEN}_3$ for graphs with two or more triangles would close the remaining cell. It is not proved in this paper; the supplement gives a conditional reduction.

**Lemma 6.3.** *Let $G$ be connected, bipartite and strong $C_4$-free, with $r(G) = 2$. Then $G$ is a tree, $\ell(G) = 2 - 2/n$ and $p(G) \le 5$.*

*Proof.* Let $s$ be a centre, and let $A$ be the colour class of $s$ and $B$ the other class. Every vertex of $B$ lies at odd distance from $s$, hence at distance 1, so $B \subseteq N(s)$. Every vertex $x \in A \setminus \{s\}$ lies at distance 2 from $s$, so $x$ has a neighbour in $B$. Two such neighbours would form a 4-cycle with $s$. Thus $x$ has exactly one neighbour. There is no edge inside $A$ or inside $B$, and the number of edges is $|B| + |A| - 1 = n - 1$. A connected graph with $n - 1$ edges is a tree. There are no triangles, so $S = 2e = 2n - 2$ and $\ell = 2 - 2/n$. Every vertex of $A \setminus \{s\}$ is a leaf. An induced path contains at most two leaves, and the unique path between them has the form leaf, neighbour in $B$, $s$, neighbour in $B$, leaf. Hence $p(G) \le 5$. □

**Lemma 6.4.** *Let $G$ be a connected triangle-free strong $C_4$-free $P_8$-free graph. Then $e(G) \le 2n-2$. In particular, for $n \ge 2$ one has $\ell(G) = 2e(G)/n < 4$.*

*Proof.* The graph has no triangle and no 4-cycle subgraph, so every cycle has length at least 5. The bound $e \le 2n - 2$ holds for $n = 1$. Let $n \ge 2$, and proceed by induction on the order. If $G$ is a tree, then $e = n - 1 \le 2n - 2$. Otherwise the girth is at least 5. Theorem 1 of Chudnovsky and Stacho [4] then asserts that $G$ is the Petersen graph, the Heawood graph, the graph obtained by contracting one edge of the Heawood graph, or else $G$ has a vertex of degree at most 2. These three exceptions have pairs $(n, e)$ equal to $(10, 15)$, $(14, 21)$ and $(13, 20)$. Each satisfies $e \le 2n-2$, since $15 \le 18$, $21 \le 26$ and $20 \le 24$.

In every remaining case some vertex $v$ has degree at most 2. Every component $H$ of $G - v$ is connected, triangle-free, strong $C_4$-free and $P_8$-free. The inductive hypothesis gives $e(H) \le 2|H| - 2$. If $G - v$ has $c \ge 1$ components, then

$$e(G) = e(G - v) + d(v) \le 2(n - 1) - 2c + 2 \le 2n - 2,$$

because $d(v) \le 2$ and $c \ge 1$. Since $G$ is triangle-free, $S = 2e \le 4n - 4$, so $\ell \le 4 - 4/n < 4$. □

Lemma 6.3 excludes the pair $(r, p) = (2, 7)$ for every connected bipartite strong $C_4$-free graph: the induced-path order is at most 5, and $\ell < 2$. Lemma 6.4 is the same exclusion for every connected triangle-free graph in the class, and it also excludes $(r, p) = (3, 7)$ from the range $5 < \ell \le 6$. A graph with $p = 7$ is $P_8$-free, so a triangle-free graph of that path order has $\ell < 4$. Thus any graph with $(r, p) \in \{(2, 7), (3, 7)\}$ and $5 < \ell \le 6$ contains a triangle. Lemma 5.9 excludes $(2, 6)$ and $(3, 6)$ for every connected strong $C_4$-free graph: $p \le 6$ forces $\ell < 4$.

The rooted bounds of orders three and four, the rooted bounds for every index, the endpoint obstructions, the high-radius constructions, and the conditional reduction for $\mathrm{GEN}_3$ are collected in the supplement. Figure 1 and Figure 2 remain in the article. The proofs of Theorems 3.6, 4.3, 5.4, 5.5 and 5.6 and of Corollary 5.11 do not use those rooted bounds or the three suppositions of

the reduction. The cut-vertex step in Theorem 5.6 repeats the comparison of Theorem A.1 and uses only Lemmas 5.3 and 2.3.

## 7. What is proved and what is open

The comparisons with Chen [3], Chen and Xu [5], Chen, Wu and Xu [6], Chudnovsky and Stacho [4], DeLaViña, Larson, Pepper and Waller [8], Karthick and Maffray [13], and Brandstädt and Hoàng [2] are in the introduction, in Section 5 and in Remarks 3.7 and 3.8. Table 1 is the only status list. An open row is not a hypothesis of a proved row. A finite search is not a proof here.

Suppose $4 < \ell \le 5$ and $G$ violates REAL-133. Then $\lceil \ell \rceil = 5$, so Corollary 6.1 with $K = 4$ gives either $r \le 4$ and $2r - 1 \le p \le r + 4$, or $r = 5$ and $p = 9$. Lemma 5.9 forces $\ell < 4$ whenever $p \le 6$, so $p \ge 7$. Corollary 4.4 forces $\ell \le 2$ when $(r, p) = (4, 7)$. The pairs that remain are $(3, 7)$, $(4, 8)$ and $(5, 9)$.

For $5 < \ell \le 6$, Corollary 6.1 gives $2r - 1 \le p \le r + 5$ and $r \le 6$. Lemma 3.1 excludes radius 1, Theorem 3.6 excludes $p \le 5$, and Corollary 4.4 excludes $(4, 7)$. Lemma 5.9 excludes $(2, 6)$ and $(3, 6)$. Lemma 6.4 shows that a triangle-free graph with $p = 7$ has $\ell < 4$, so the pairs $(2, 7)$ and $(3, 7)$ are possible in this range only for graphs that contain a triangle. Lemma 6.3 is the stronger bipartite exclusion of $(2, 7)$. The pairs that remain are necessary conditions for a counterexample. The table does not assert that any of them occurs.

No structure theorem for connected strong $C_4$-free graphs of radius $r \ge 5$ and path order $2r - 1$ is proved here. Notes on deletion in the triangle-free class, and on ports of a distance-preserving dominating set, are kept outside this article in `notes/supplement-deletion-ports.tex`.

**Conjecture 7.1.** ROOT($q$) *holds for every $q \ge 3$. For every $r \ge 5$, every connected strong $C_4$-free graph of radius $r$ and path order $2r - 1$ satisfies $S \le (r - 1)n - 2r(r - 3)$, with equality only for $C_{2r}$.*

The cases $q = 3$ and $q = 4$ are Lemma 5.3 and Theorem 1.2, and every $q$ holds in $\mathcal{B}$ by Theorem 2.2. The radius bound is not claimed for $r \ge 5$.

**Conjecture 7.2.** *If $F$ is connected and strong $C_4$-free, $p_s(F) = q$ and the eccentricity of $s$ is $q - 1$, then*

$$S(F) \le (q - 1)(n - 1) - (q - 3)D - m - (q - 2)(q - 3).$$

For $q = 4$ the subtracted term is 2, and Proposition 1.3 is this case. For $q \ge 5$ the conjecture is open. The terminal bound in $\mathcal{B}$ is the first part of Theorem 2.2.

Conjecture 7.2, for a fixed $q$, implies ROOT($q$). The implication does not prove the conjecture. Let $F$ be connected and strong $C_4$-free with $p_s(F) \le q$. Form $G$ by attaching a path of $q - 1$ new edges at $s$. Then $G$ is terminal of index $q$: the new path is an induced path of order $q$ starting at $s$, and every induced path of $F$ that starts at $s$ has order at most $q$. The order, the root degree and the mass increase by $q - 1$, 1 and $2(q - 1)$ respectively, and $m$ is unchanged. Write $N = n + q - 1$ and $D_G = D + 1$. The terminal inequality in $G$ reads

$$S(F) + 2(q - 1) \le (q - 1)(N - 1) - (q - 3)D_G - m - (q - 2)(q - 3).$$

The identity $N - 1 = n + q - 2$ gives $(q - 1)(N - 1) = (q - 1)(n - 1) + (q - 1)^2$, and $(q - 3)D_G = (q - 3)D + (q - 3)$. The terms that do not involve $n$, $D$ or $m$ are

$$(q - 1)^2 - (q - 3) - (q - 2)(q - 3) - 2(q - 1) = (q - 3)\big((q - 1) - (q - 1)\big) = 0.$$

What remains is $S(F) \le (q - 1)(n - 1) - (q - 3)D - m$, which is ROOT($q$). For $q = 4$ this route uses Proposition 1.3. The induction in Theorem 1.2 does not. For $q \ge 5$ the hypothesis is open, so the implication leaves ROOT($q$) open.

| Statement | Status |
|---|---|
| $M(n)$ for connected strong $C_4$-free $P_6$-free graphs | proved, Theorem 3.6 |
| the structure of graphs with $r = 4$ and $p = 7$, the formula $S = 2n - b - m$, and the equality family | proved, Theorem 4.3 and Corollaries 4.4, 4.5 |
| $2e \le 3n$ for connected triangle-free strong $C_4$-free $P_7$-free graphs, equality only for the Petersen graph | proved, Theorem 5.4 |
| the exact value $f(n)$ for the same class, $n \ge 10$ | proved, Theorem 5.5 |
| $S \le 3n$ for connected strong $C_4$-free $P_7$-free graphs with at most one triangle | proved, Theorem 5.6 |
| $\Delta \ge 1$ for $n \ge 4$, hence $e \le 2n - 4$ and $\ell < 4$; if $n \ge 6$ and $T \le 2$, then $\Delta \ge 2$ | proved, Lemmas 5.9 and 5.10 |
| REAL-133 for $\ell \le 3$; for $r \le 2$ and $\ell \le 4$; for $r = 4$ and $\ell \le 4$; for $r \ge 5$ and $\ell \le 4$ | proved, Theorem 5.13 |
| ROOT(3) and ROOT(4); the terminal strengthening of Proposition 1.3 | proved in the supplement |
| ROOT($q$) for every $q$, in the class $\mathcal{B}$ | proved in the supplement, Theorem 2.2 |
| a minimum counterexample to $\mathrm{GEN}_4$ is 2-connected | proved in the supplement, Proposition 1.4; $\mathrm{GEN}_4$ is open |
| the cell $r = 3$, $p = 6$, $T \ge 2$, $n \ge 17$, $3 < \ell \le 4 - 16/n$ | open; Corollary 5.11, and $\ell = 4$ does not occur |
| $\mathrm{GEN}_3$ for graphs with two or more triangles | open; the reduction in the supplement is conditional |
| $\mathrm{GEN}_K$ for $K \ge 4$, unrestricted $\mathrm{GEN}_6$, RAD($r$) for $r \ge 5$, and ROOT($q$) for $q \ge 5$ outside $\mathcal{B}$ | open |
| REAL-133 | open |
| pairs $(r, p)$ still necessary for $4 < \ell \le 5$ | $(3, 7)$, $(4, 8)$, $(5, 9)$ |
| pairs $(r, p)$ still necessary for $5 < \ell \le 6$ | $(2, 7)$, $(3, 7)$, $(3, 8)$, $(4, 8)$, $(4, 9)$, $(5, 9)$, $(5, 10)$, $(6, 11)$ |
| pairs excluded from $5 < \ell \le 6$ for every strong $C_4$-free graph | $(2, 6)$ and $(3, 6)$, by Lemma 5.9 |
| $(2, 7)$ and $(3, 7)$ in the range $5 < \ell \le 6$ | each such graph contains a triangle, by Lemma 6.4 |

TABLE 1. Proved statements and statements left open. The rows for $4 < \ell \le 5$ and for $5 < \ell \le 6$ record necessary pairs only.

**Conjecture 7.3.** *Every connected strong $C_4$-free graph $G$ that is not a cycle and has $\ell(G) > 5$ has an eligible core $H$, in the sense of Example 3.5, with*

$$S(G) - S(H) \le (p - r)\big(n - |H|\big) + |H|.$$

Example 3.5 shows that the budget without a restriction on $\ell$ is false, and that the Petersen graph is a counterexample with $\ell = 3$. The example does not show that 5 is the correct cutoff, and it does not count eligible cores. A reported count of cores of order at most 12 is not a result of this paper. If the conjecture, the range $\ell \le 5$, and an induction on the order were all available, REAL-133 would follow. The range $\ell \le 5$ is not a theorem, so this route is not a proof of REAL-133. Example 3.6 shows that a proof cannot rely only on the scalar constraints listed there.

## Acknowledgments

The authors are responsible for the text, including any errors. Language-model tools were used to prepare drafts. Neil Xu's correspondence address is omitted from this version. Xingchen Wang is listed by name only.

## References

[1] E. Camby and O. Schaudt, *A new characterization of $P_k$-free graphs*, Algorithmica **75** (2016), 205–217.
[2] A. Brandstädt and C. T. Hoàng, *On clique separators, nearly chordal graphs, and the Maximum Weight Stable Set Problem*, Theoret. Comput. Sci. **389** (2007), 295–306.
[3] H. Chen, *Partial results on the WOWII-133 Graffiti conjecture*, Zenodo, 2026, `https://doi.org/10.5281/zenodo.22054834` (version record 22054835).
[4] M. Chudnovsky and J. Stacho, *3-colorable subclasses of $P_8$-free graphs*, SIAM J. Discrete Math. **32** (2018), 1111–1138.
[5] R. Chen and B. Xu, *Nearly optimal coloring of some $C_4$-free graphs*, Discrete Appl. Math. (2026), `https://doi.org/10.1016/j.dam.2026.07.010`; arXiv:2409.06944.
[6] R. Chen, D. Wu and B. Xu, *Structure of some $(P_7, C_4)$-free graphs with application to colorings*, Discrete Appl. Math. **357** (2024), 14–23; arXiv:2302.11115.
[7] E. DeLaVina, *Written on the Wall II (Conjectures of Graffiti.pc)*, `http://cms.dt.uh.edu/faculty/delavinae/research/wowII/`.
[8] E. DeLaViña, C. E. Larson, R. Pepper and B. Waller, *A characterization of graphs where the independence number equals the radius*, Graphs Combin. **28** (2012), 315–332.
[9] S. Fajtlowicz, *A characterization of radius-critical graphs*, J. Graph Theory **12** (1988), 529–532.
[10] P. Erdős, M. Saks and V. T. Sós, *Maximum induced trees in graphs*, J. Combin. Theory Ser. B **41** (1986), 61–79.
[11] Formal Conjectures (Google DeepMind), file `FormalConjectures/WrittenOnTheWallII/GraphConjecture133.lean` at commit `5faef0c4dc3d226f8f9a9fc8f581b6221283eb6e`, `https://github.com/google-deepmind/formal-conjectures/blob/5faef0c4dc3d226f8f9a9fc8f581b6221283eb6e/FormalConjectures/WrittenOnTheWallII/GraphConjecture133.lean`.
[12] Formal Conjectures, pull request #5727, merged as commit `5faef0c4dc3d226f8f9a9fc8f581b6221283eb6e`, `https://github.com/google-deepmind/formal-conjectures/pull/5727`.
[13] T. Karthick and F. Maffray, *Square-free graphs with no six-vertex induced path*, SIAM J. Discrete Math. **33** (2019), 874–909; arXiv:1805.05007.
[14] D. Seinsche, *On a property of the class of $n$-colorable graphs*, J. Combin. Theory Ser. B **16** (1974), 191–193.